

National Institute of Business Management

Colombo – 07

School of Computing and Engineering

DIPLOMA IN SOFTWARE ENGINEERING

Final Project Report Documentation

24.3F

Submission Date – 20/07/2026

Project Title: Website Development for Paradyze (PVT) Limited – An Online Clothing Store

Group Members:

1. S.N.P. Paranagama CODSE24.3F-001
2. W.H.M.V.D. Bandara CODSE24.3F-035
3. R.P. Wickramasekara CODSE24.3F-036
4. J.A.D. Anjani CODSE24.3F-042

Name of the Program: Diploma in Software Engineering

Institution: School of Computing and Engineering, National Institute of Business Management

Date: 20/07/2026

Declaration

We certify that this project does not incorporate without acknowledgement, any material previously submitted for a Diploma in any institution and to the best of my knowledge and belief it does not contain any material previously published or written by another person or myself except where due reference is made in the text. I also hereby give consent for my project report, if accepted, to be made available for photocopying and for interlibrary loans, and for the title and summary to be made available to outside organizations.

Signatures:

S.N.P. Paranagama	……	Date …….
W.H.M.V.D. Bandara	……	Date …….
R.P. Wickramasekara	...…….	Date …….
J.A.D. Anjani	…….	Date …….

Abstract

Paradyze (Private) Limited is a Sri Lankan online clothing business that primarily promotes and sells fashion products through social media platforms such as Instagram and Facebook. Its existing operations depend heavily on direct messages, spreadsheets and manual communication to process orders, maintain stock information, confirm payments and arrange deliveries. These methods can result in inventory inaccuracies, delayed order processing, limited customer information management and difficulty generating reliable business reports.

This project focused on the analysis, design and development of a centralized Online Clothing Store Management System for Paradyze. The proposed system provides customers with facilities to register accounts, log in securely, browse and search for products, manage shopping carts and wishlists, place orders, view order histories, track order statuses and submit product reviews. It also provides staff members with role-based access to manage products, inventory, customers, orders, suppliers, deliveries, returns, refunds and reports.

Agile software development practices were followed to support iterative development, regular testing and continuous improvements. Interviews, observations, questionnaires and document reviews were used to identify the requirements of the organization and its customers. UML diagrams, sequence diagrams and an entity relationship diagram were used to model the proposed system before implementation.

The developed solution is expected to reduce manual work, improve inventory accuracy, provide better access to business information and enhance the overall customer shopping experience. It also provides a scalable foundation for future improvements such as online payment integration, courier tracking, cloud deployment and advanced sales analytics.

Keywords

Online Clothing Store E-Commerce, Inventory Management, Order Management, Customer Management, Role-Based Access Control, Product Management, Supplier Management, Return and Refund Management, Sales Reporting, Web Application, Database Management System, Paradyze

Table of Content

Declaration	03
Abstract	04
Keywords	04
Acknowledgment	09
Chapter 1: Introduction	10
1.1 Introduction of the Organization	10
1.2 Organizational Structure	11
1.3 Current Operations in Organization	12
1.4 Users and Responsibilities in Organization	13
1.5 Problem Definition	15
1.6 Project Objectives	16
1.7 Proposed Solution	17
1.8 Chapter Summary	20
Chapter 2: Methodology	21
2.1 Introduction	21
2.2 Data Collection Methods	22
2.3 Software Process Model	23
2.4 Software Development Tools	25
2.5 Testing Strategies	27
2.6 Implementation Plan	29
2.7 Chapter Summary	30
Chapter 3: Analysis	31
3.1 Introduction	31
3.2 UML Diagrams	32
3.3 ER Diagram of the Proposed System	40
3.4 Chapter Summary	41
Chapter 4: Solution Design	42
4.1 Introduction	42
4.2 Interface Design	43
4.3 Database Design	61
4.4 Report Layout Design	63
Chapter 5: Conclusion	64

List of Figures

Figure 1.1 12

Figure 2.1 25
Figure 2.2 26
Figure 2.3 26
Figure 2.4 26
Figure 2.5 26
Figure 2.6 29
Figure 2.7 27
Figure 2.8 27
Figure 2.9 27
Figure 2.10 28
Figure 2.11 29

Figure 3.1 35
Figure 3.2 36
Figure 3.3 37
Figure 3.4 37
Figure 3.5 37
Figure 3.6 38
Figure 3.7 38
Figure 3.8 38
Figure 3.9 38
Figure 3.10 38
Figure 3.11 39
Figure 3.12 39
Figure 3.13 39
Figure 3.14 39
Figure 3.15 39
Figure 3.16 40
Figure 3.17 40
Figure 3.18 40
Figure 3.19 40
Figure 3.20 41

Figure 4.1 45
Figure 4.2 46
Figure 4.3 47
Figure 4.4 48
Figure 4.5 49

Figure 4.6 50
Figure 4.7 51
Figure 4.8 52
Figure 4.9 53
Figure 4.10 54
Figure 4.11 55
Figure 4.12 56
Figure 4.13 57
Figure 4.14 58
Figure 4.15 59
Figure 4.16 60
Figure 4.17 61
Figure 4.16 64

List of Tables

Table 1 09

Table 2 63

List of Acronyms and Abbreviations

Acronym	Meaning
API	Application Programming Interface
CSS	Cascading Style Sheets
CRUD	Create, Read, Update and Delete
DB	Database
DBMS	Database Management System
DSE	Diploma in Software Engineering
ER	Entity Relationship
ERD	Entity Relationship Diagram
FK	Foreign Key
HTML	Hyper Text Markup Language
HTTP	Hyper Text Transfer Protocol
ID	Identifier
MIS	Management Information Systems
NIBM	National Institute of Business Management
PDF	Portable Document Format
PK	Primary Key
PVT	Private
RBAC	Role-Based Access Control
SQL	Structured Query Language
UI	User Interface
UML	Unified Modeling Language
URL	Uniform Resource Locator
UAT	User Acceptance Testing
UX	User Experience
WWW	World Wide Web
JS	JavaScript
OTP	One-Time Password
CSV	Comma-Separated Values
SMS	Short Message Service
SEO	Search Engine Optimization
SSL	Secure Sockets Layer
TLS	Transport Layer Security

Table 1

Acknowledgement

We would like to express our sincere gratitude to everyone who contributed to the successful completion of this project.

First, we would like to thank the National Institute of Business Management and the School of Computing and Engineering for providing us with the knowledge, resources and opportunity required to undertake this project as part of the Diploma in Software Engineering program.

We extend our deepest appreciation to our project supervisor, Ms.Chandula, for the continuous guidance, valuable suggestions, constructive feedback and encouragement provided throughout the analysis, design, development, testing and documentation stages of the project.

We would also like to express our gratitude to the management and staff of Paradyze (Private) Limited for granting us permission to conduct the project within the organization. Their cooperation in providing information about existing business activities, order-processing procedures, inventory management and customer-service operations was essential for identifying the system requirements.

Our appreciation is also extended to the customers and other participants who responded to interviews and questionnaires and provided valuable feedback regarding their expectations of an online clothing store.

We would also like to thank all lecturers of the Diploma in Software Engineering program for the knowledge and technical guidance provided throughout our studies.

Finally, we express our heartfelt gratitude to our families, friends and group members for their patience, encouragement, cooperation and support throughout the completion of this project.

Chapter 1: Introduction

1.1 Introduction of the Organization

Paradyze (PVT) Limited is a startup in Sri Lanka aiming to redefine the clothing retail experience by operating as a fully online fashion store. With a vision to offer trendy, affordable, and quality fashion items, Paradyze primarily targets young adults and teenagers.

- **Vision:** To become Sri Lanka's most trusted and stylish online fashion destination.
- **Mission:** To provide affordable and fashionable clothing through a fast, reliable, and user-friendly e-commerce website

1.2 Organizational Structure

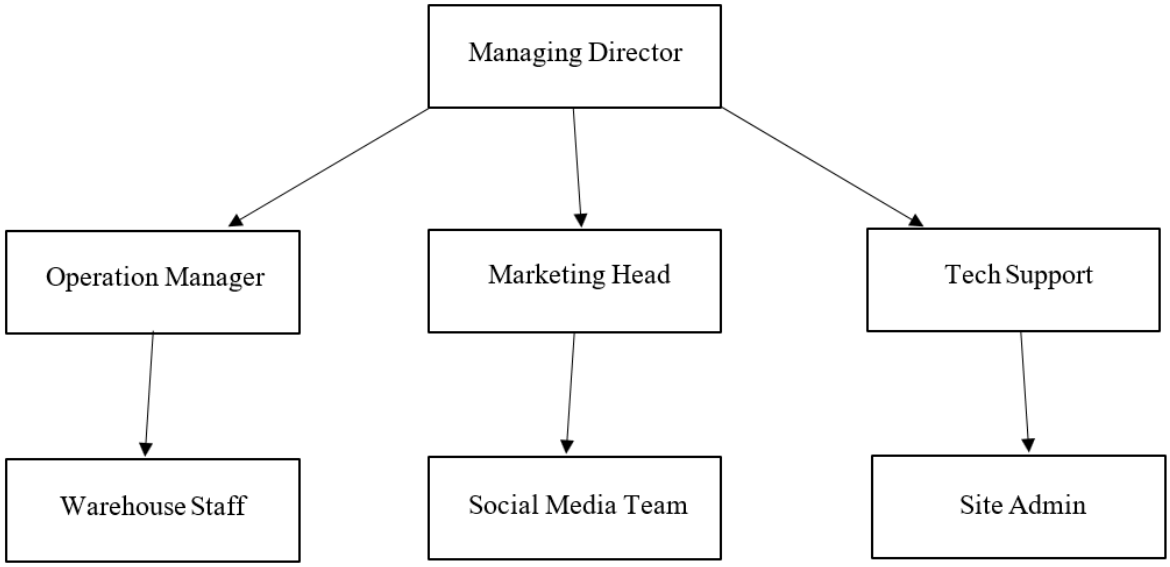


Figure 1.1

1.3 Current Operations in Organization

Paradyze currently runs its business through social media platforms, mainly Instagram and WhatsApp. Products are showcased through posts and stories, and customers place orders by sending direct messages.

All orders are managed manually using spreadsheets, and stock levels are tracked by staff based on incoming messages and product availability. There is no real-time stock monitoring, leading to frequent inventory errors and overselling.

Payments are via bank transfers or mobile apps and order fulfillment is handled using third party delivery services, coordinated manually. Customer records are not stored systematically and the business lacks automation for order tracking, report generation or performance monitoring.

As order volume increases, this manual approach causes inefficiencies and limits the potential for growth.

1.4 Users and Responsibilities in Organization

Managing Director

- Oversees overall business strategy and growth.
- Approves major decisions related to sales, marketing, and operations.
- Monitors performance through system reports.
- Delegates tasks and evaluates departmental performance.

Operation Manager

- Manages day-to-day business operations.
- Oversees order fulfillment and logistics.
- Coordinates between departments for smooth workflow.
- Reviews operational reports and update management.

Marketing Manager

- Manages product promotions and online advertising.
- Plans and executes seasonal campaigns and discounts.
- Uploads product banners, offers, and announcements on the website.
- Tracks performance of campaign via analytics.

Store Manager

- Maintains and updates product inventory in the system.
- Supervises warehouse staff and order packaging.
- Monitors low-stock items and generates restocking alerts.
- Ensures timely dispatch of confirmed orders.

Warehouse Staff

- Picks and packs items based on confirmed orders.
- Verifies the quality before dispatching.
- Assists in stock audits and physical inventory checks.

Social Media Team

- Posts new product collections and campaigns online.
- Manages daily engagement on Instagram and WhatsApp.
- Handles basic customer questions via DMs before redirecting to support.

Customers

- Responds to customer inquiries via chat/email.
- Assists with order tracking, returns, and refunds.
- Manages complaints and escalates issues to managers.
- Provides feedback to improve service quality.

1.5 Problem Definition

Paradyze currently relies on manual processes and social media platforms to run its online clothing business. While this setup has supported the business in its early stages, it now presents several challenges as customer demand and product variety grow. Orders are taken through direct messages and tracked using spreadsheets, which often leads to human errors, miscommunication, and delayed order processing.

Inventory management is another major issue with no real-time system to monitor stock levels. This results in overselling, out-of-stock issues and customer dissatisfaction. Payments and delivery arrangements are handled separately and manually, causing inefficiencies and delays. Additionally, there is no central database to track customer information, order history, or sales analytics limiting the company's ability to make data-driven decisions.

As Paradyze scales up, the lack of an integrated management system affects its operational efficiency, customer experience, and overall growth potential. To address these challenges, a centralized Online Clothing Store Management System is essential for streamlining operations, automating key processes and supporting future expansion.

1.6 Project Objectives

- **Create a Polished, mobile-friendly e-commerce website.**

  To give customers a dependable and convenient shopping experience, create a simple, aesthetically pleasing website that functions flawlessly on PCs, tablets and smartphones.

- **Facilitate secure checkout, product search and customer registration.**

  Using a safe and easy checkout process, let customers register, browse or search products, add items to their cart and finish purchases.

- **Offer an administrative dashboard for stock and order management.**

  Give administrators a centralized dashboard so they can effectively view customer data, add or remove products, track orders in real time, and update inventory levels.

- **A neat, branded user interface will enhance the client experience.**

  Create a user interface that embodies the Paradyze brand while guaranteeing sample navigation, eye-catching product displays and seamless interaction for customers.

- **Automate sales reports, order tracking and inventory updates.**

  Integrate features that automatically generate sales reports, update stock levels after each purchase, and allow customers to track their order status without manual input.

1.7 Proposed Solution

Our development team suggests creating and implementing a centralized online clothing management system to help Paradyze its online clothing business. With back-end management features integrated, this system will function as a fully functional e-commerce platform.

Every step of the business process will be streamlined by the solution, including inventory control, order management, report generation, product listing and customer interaction. Modern, user-friendly technologies will be used in the system's development to guarantee scalability, security and dependability. Through a dedicated back-end interface, administrators and employees will be able to effectively manage business operations while customers will be able to interact with the store via a responsive website.

Functional Requirements:

- **Customer Management Module**
  - User registration and profile management
  - Login/Logout with password reset
  - Order history and tracking
  - Wishlist and review features

- **Product and Inventory Module**
  - Product listing with categories, size and filter options
  - Real-time stock update
  - Low-stock alerts for admin
  - Automated inventory deduction on sale
  - Add product variants
  - Product zoom and image gallery
  - Product size guide popup
  - Admin alerts when stock is low

- **Order Management Module**
  - Add to cart and secure checkout
  - Order confirmation, status update and invoice generation
  - Return/refund request handling
  - Integration with payment gateways
  - Order cancellation request (before dispatch)

- **Admin Dashboard**
  - View total orders, active customers, top-selling products
  - Add/edit/delete products and categories
  - Manage inventory levels and restocking
  - Approve returns and view sales statistics
  - View customer order history

- **Report Generation**
  - Daily/weekly/monthly sales reports
  - Inventory summary reports
  - Customer activity logs
  - Export reports as PDF or Excel

- **Customer Communication Module**
  - Contact form (inquires go to admin email)
  - Email/SMS notifications for orders, status updates

- **Shipping and Delivery Management**
  - Estimated delivery time display
  - Order delivery tracking integration
  - Printable order invoices and delivery slips

- **Supplier Management Module**
  - Add/edit supplier profiles with contact and product info
  - Generate purchase orders for restocking
  - Track supplier deliveries and restock status
  - View supplier order history and performance

- **Employee Management Module**
  - Role based access
  - Staff login with dashboard access based on role
  - Activity logs for user actions
  - Internal system alerts to staff

Non-Functional Requirements:

- **Security**
  - Role-based access control
  - Encryption for secure transactions
  - Password hashing and user data protection

- **Usability**
  - Clean UI for both customers and staff
  - Fully responsive across all devices

- **Performance**
  - Fast loading time for product pages
  - Optimized image handling
  - Real-time database syncing

- **Backup and Recovery**
  - Daily automated backups
  - Quick restore options in case of system failure

The goal of this solution is to streamline and digitize Paradyze's entire business process, increasing efficiency, accuracy and customer satisfaction. Paradyze will be able to expand its business, provide a polished shopping experience and maintain its competitiveness in the online fashion retail sector with the implementation for suggested system.

1.8 Chapter Summary

This chapter gives a thorough rundown of Paradyze's current business operations and the main issues the company is facing as a result of its manual and disjointed processes. It pinpointed particular issues with order management, inventory tracking, customer service and general operational effectiveness.

The chapter addressed these problems by outlining the project goals, which included transforming the company with a centralized, easy navigable Online Clothing Store Management System. The functional and non-functional elements that will solve current constraints and enhance business performance were highlighted in the comprehensive suggested solution.

It is anticipated that the suggested system will improve customer satisfaction, decrease manual labor and streamline operations, eventually allowing Paradyze to grow and successfully compete in the online retail market.

Chapter 2: Methodology

2.1 Introduction

The approach that will be used in the creation of Paradyze's online clothing store management system is described in this chapter. It outlines the steps the project team will take to collect requirements, schedule the development process, select appropriate tools and technologies and test the system to guarantee dependability and functionality.

While providing flexibility for enhancements throughout the development cycle, the selected methodology is intended to guarantee that the finished product fulfils the expectations of the company and its clients. The team wants to provide a safe, scalable and intuitive solution that tackles all of the business issues that have been identified by using an organized and iterative process.

2.2 Data Collection Methods

Several data collection techniques were employed to obtain precise and pertinent information from the stakeholders in order to guarantee that the suggested Online Clothing Store Management System satisfies Paradyze's needs. By using these techniques, the development team was able to comprehend current processes, pinpoint major issues and ascertain the system's functional and non-functional requirements.

- **Interviews**

  One-on-one discussions were held with the business owner, warehouse staff and customer support personnel to understand their roles, challenges, and expectations from the new system. These interviews provided detailed insights into current order handling, inventory issues and customer service limitations.

- **Observations**

  The development team observed the day-to-day operations of Paradyze, including how orders were received via social media, how spreadsheets were used to manually manage inventory, and how delivery coordination was handled. As a result, the group was able to pinpoint inefficiencies and potential areas for automation.

- **Questionaries**

  Regular customers were given a brief set of structured questions to complete in order to get their opinions on the shopping experience. Customer expectations for product search, checkout, payment options and order tracking were ascertained through the responses.

- **Document Analysis**

  To learn more about how the company manages inventory, handles payments, and interacts with clients, existing documents including sales spreadsheets, order logs and social media exchanges were examined.

All of these techniques combined to assist the team in obtaining the data required to create a system that complements Paradyze's ongoing business practices while incorporating enhancements that boost productivity, scalability and client satisfaction.

2.3 Software Development Model

Our development team has decided that the Agile Software Development Model is the best method for creating the Online Clothing Store Management System because of the dynamic nature of e-commerce and the changing needs of a developing company like Paradyze.

Agile provides an adaptable and iterative framework that facilitates frequent delivery of functional features, client collaboration and ongoing improvement. This model works especially well for projects where system requirements may change over time and user input is critical.

Why Agile is suitable for this project?

- **Flexibility and Iterative Development**

  The system can be developed in incremental stages, allowing the team to release usable features early, gathers feedback and make improvements continuously.

- **Client Involvement**

  Regular feedback from Paradyze's staff and potential users ensures that the final product closely aligns with real business needs and user expectations.

- **Risk Reduction**

  Issues can be identified and resolved at early stages in the development cycle, reducing the chances of failure at the final delivery stage.

- **Improved Collaboration**

  Agile promotes constant communication between team members and stakeholders, ensuring transparency, accountability and adaptability throughout the project.

- **Continuous Testing and Delivery**

  Each iteration includes testing and review, which helps maintain high software quality and ensures that functionality is reliable and ready to deploy.

By following the Agile Model, the development team will be able to deliver a high-quality, user-focused system in a structured yet flexible manner, ensuring it meets the operational needs of Paradyze while maintaining room for scalability and enhancements in future iterations

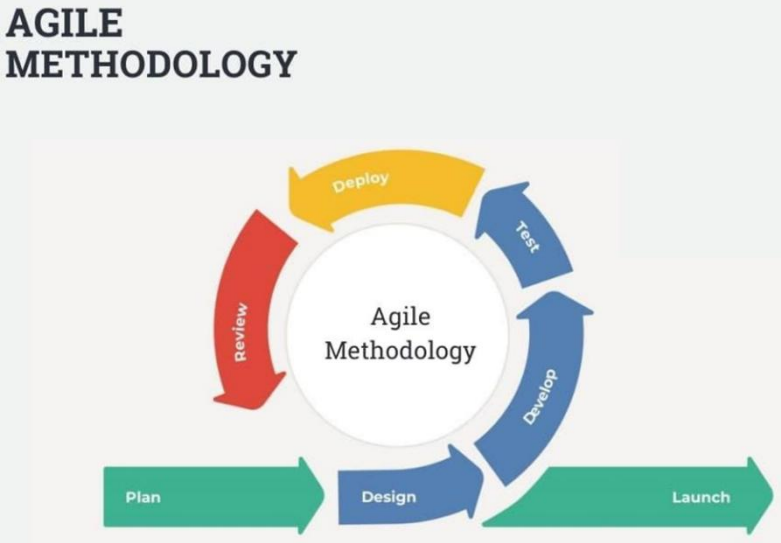

AGILE
METHODOLOGY
Deploy
Test
Review
Agile
Methodology
Develop
Plan
Design
Launch

Figure 2.1

2.4 Software Development Tools

The project team will make use of a number of cutting-edge and dependable tools and technologies in order to successfully design, develop and implement the Online Clothing Store Management System for Paradyze. These were chosen due to their scalability, community support, ease of use, compatibility with the project scope and suitability for e-commerce applications.

- **VS Code:** Write the code to debug and deployment.



Figure 2.2

- **MySQL:** Store and manage all dynamic data.



Figure 2.3

- **Figma:** Create UI/UX wireframes and prototypes.



Figure 2.4

- **Wordfence Security Plugin:** Ensures protection against common threats.



Figure 2.5

- **UpdraftPlus:** Automates daily backups.



Figure 2.6

- **Rank Math:** To improve visibility and enhance user reach.



Figure 2.7

- **Smush:** Used for image optimization.



Figure 2.8

- **GitHub:** Maintains version history of the project.



Figure 2.9

2.5 Testing Strategies

To make sure the Online Clothing Store Management System is dependable, safe and operates as intended, testing is an integral part of the development process. To thoroughly assess the system and guarantee its deployment readiness, the development team will combine Black Box and White Box testing methodologies.

- Black Box Testing

  This focuses on evaluating the system's functionality from the user's perspective, without knowledge of the internal workings. The goal is to test whether the system behaves as expected under real-world conditions.

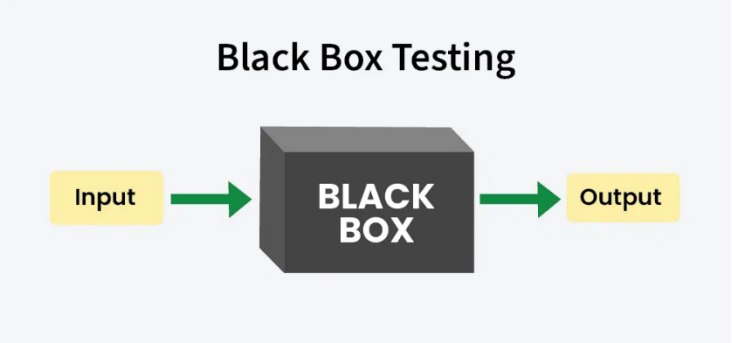


Figure 2.10

Functional Testing: This will involve testing each function of the system.

Usability Testing: The system's UI will be tested for ease for use.

Performance Testing: This will be tested under various load conditions to check how it handles multiple users, high traffic and heavy data processing, ensuring optimal performance without delays.

Security Testing: This will focus on ensuring system's security measures.

- White Box Testing

This involves evaluating the internal workings of the system, including the source code, database interactions and logic flows.

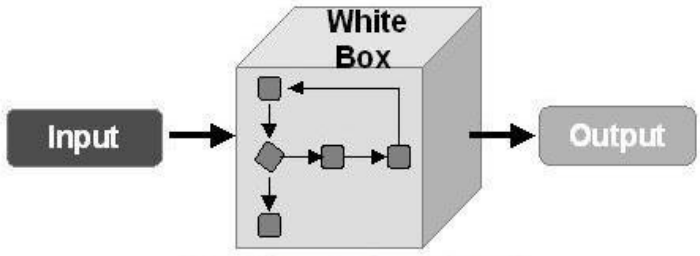


Figure 2.11

Unit Testing: Each module will be individually tested to ensure that all individual components work as expected.

Integration Testing: This will test how well the different modules of the system work together.

Database Testing: Testing will ensure that the MySQL database interacts correctly with the front-end system.

Regression Testing: After each update or change to the system, this will be performed to ensure that new features do not break existing functionality.

By using these testing strategies, the team will ensure that the Online Clothing Store Management System is vetted for functionality, security and performance for Paradyze's growth.

2.6 Implementation Plan

The implementation plan offers a thorough road map for the effective creation and rollout of Paradyze's online clothing store management system. This strategy describes the project's methodical development from the first requirement collection to the last delivery, guaranteeing that every phase is finished on schedule and significantly advances the system's overall goals.

The Agile methodology, which encourages adaptability, ongoing feedback and incremental delivery will be used by the development team. Each weekly phase of the project will concentrate on a different aspect of the system, such as feature development, back-end configuration, user interface design, testing and deployment.

This structured approach allows the team to :

- Identify and resolve issues early
- Adapt to evolving requirements
- Maintain steady progress
- Ensure high system quality and user satisfaction

To verify system performance, security and functionality, frequent testing and stakeholder reviews will be carried out during the implementation. In order to guarantee the platform's long-term dependability, the plan also incorporates documentation, optimization and backup plans.

Following this plan will help the development team deliver a fully functional, secure and user-friendly management system that satisfies Paradyze's operational requirement and facilities future business growth.

2.7 Chapter Summary

The approach used to create Paradyze's online clothing store management system was described in this chapter. It started by outlining how crucial it is to employ an organized strategy in order to guarantee the project's successful completion.

To demonstrate how the system requirements were precisely obtained, a variety of data collection techniques were covered, including observations, questionnaires, interviews and document analysis. Because of its adaptability, iterative nature and responsiveness to feedback, the Agile Software Development Model was determined to be the best strategy for this project.

Additionally, the chapter introduced the technologies and tools that were chosen for the purpose of creating a scalable and user-friendly system. These included WordPress, WooCommerce, Elementor, MySQL, Figma and security plugins. In order to guarantee the system's dependability, security and usability, testing strategies encompassing both black box and white box techniques were examined.

A thorough implementation plan that broke down the tasks week by week and explain how each stage adds to the overall development process was finally presented. The team will follow the tactics covered in this chapter to deliver a comprehensive, useful and effective management system that satisfies Paradyze's operational requirements.

Chapter 3: Analysis

3.1 Analysis

System analysis is the process of examining the existing business environment, identifying its weaknesses and determining the functional and data requirements of the proposed solution. This chapter presents the analysis conducted for the proposed Paradyze Online Clothing Store Management System.

Paradyze (Private) Limited currently conducts most of its sales activities through Instagram and WhatsApp. Customers view products through social-media posts and communicate with staff using direct messages. Orders, payments, customer details and inventory information are maintained manually using messages and spreadsheets. This creates difficulties such as inaccurate stock information, delayed order processing, duplicate records, poor order tracking and limited access to business reports.

The proposed system was analyzed using Unified Modeling Language diagrams and an Entity Relationship Diagram. The UML diagrams demonstrate the behavior and structure of the proposed solution, while the ER diagram represents the data entities, attributes, relationships and cardinalities required by the database.

The following analysis models are presented in this chapter:

- Use Case Diagrams of the Proposed System
- Class Diagram of the Proposed System
- Sequence Diagrams for the Proposed System
- Entity Relationship Diagram of the Proposed System

3.2 UML Diagrams

Unified Modeling Language, commonly known as UML, is a standard visual modelling language used to analyze, design and document software systems. UML diagrams allow developers, users and other stakeholders to understand the structure and behavior of a system before and during implementation.

Behavioral UML diagrams, such as use case and sequence diagrams, show how users interact with the system and how system components communicate with each other. Structural UML diagrams, such as the class diagram, show the classes, attributes, operations and relationships within the system.

For the Paradyze project, UML diagrams were used to:

- Understand the existing manual business processes.
- Identify the users of the proposed system.
- Define the functions available to each user.
- Analyse interactions between users and system components.
- Identify major system classes and their relationships.
- Reduce ambiguity during system development.
- Provide a common reference for developers and stakeholders.

3.2.1 Use Case Diagram of the Proposed System

The proposed system use case diagram represents the interactions between customers, staff members, managers and external services within the computerized Paradyze Online Clothing Store Management System.

The proposed solution introduces a centralized web-based platform through which customers can browse products, create accounts, manage wishlists, add products to a cart, place orders, make payments, track deliveries, review products and request returns or refunds.

Staff members access a separate dashboard. The functions visible to a staff member depend on the assigned role. The principal staff roles are Administrator, Operations Manager, Marketing Manager, Store Manager and Warehouse Staff. Role-based access prevents users from accessing functions that are outside their responsibilities.

The proposed system also communicates with external actors such as payment services, suppliers and delivery services.

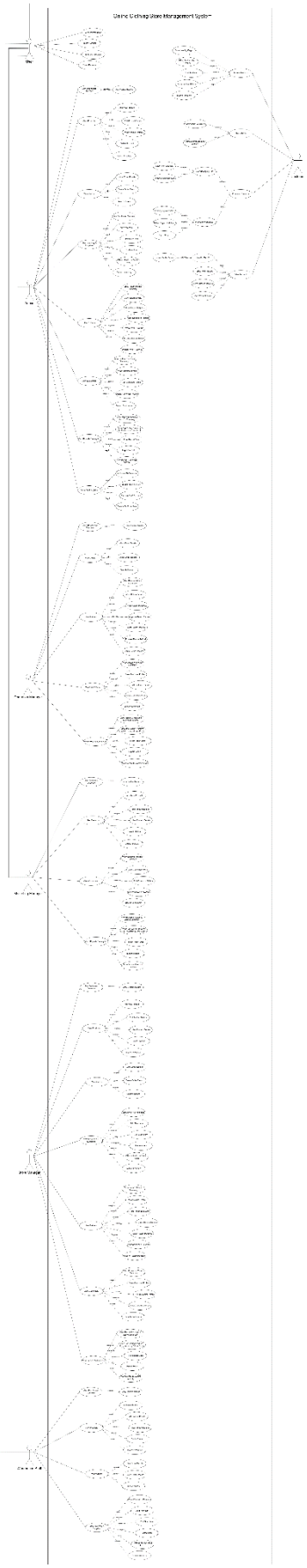


Figure 3.1

3.2.2 Class Diagram of the Proposed System

The class diagram represents the static structure of the proposed Paradyze system. It identifies the main classes, their attributes, operations and relationships.

The diagram supports object-oriented analysis by showing how customer accounts, staff accounts, products, inventory, carts, orders, payments, shipping, suppliers, returns and other system components are connected.

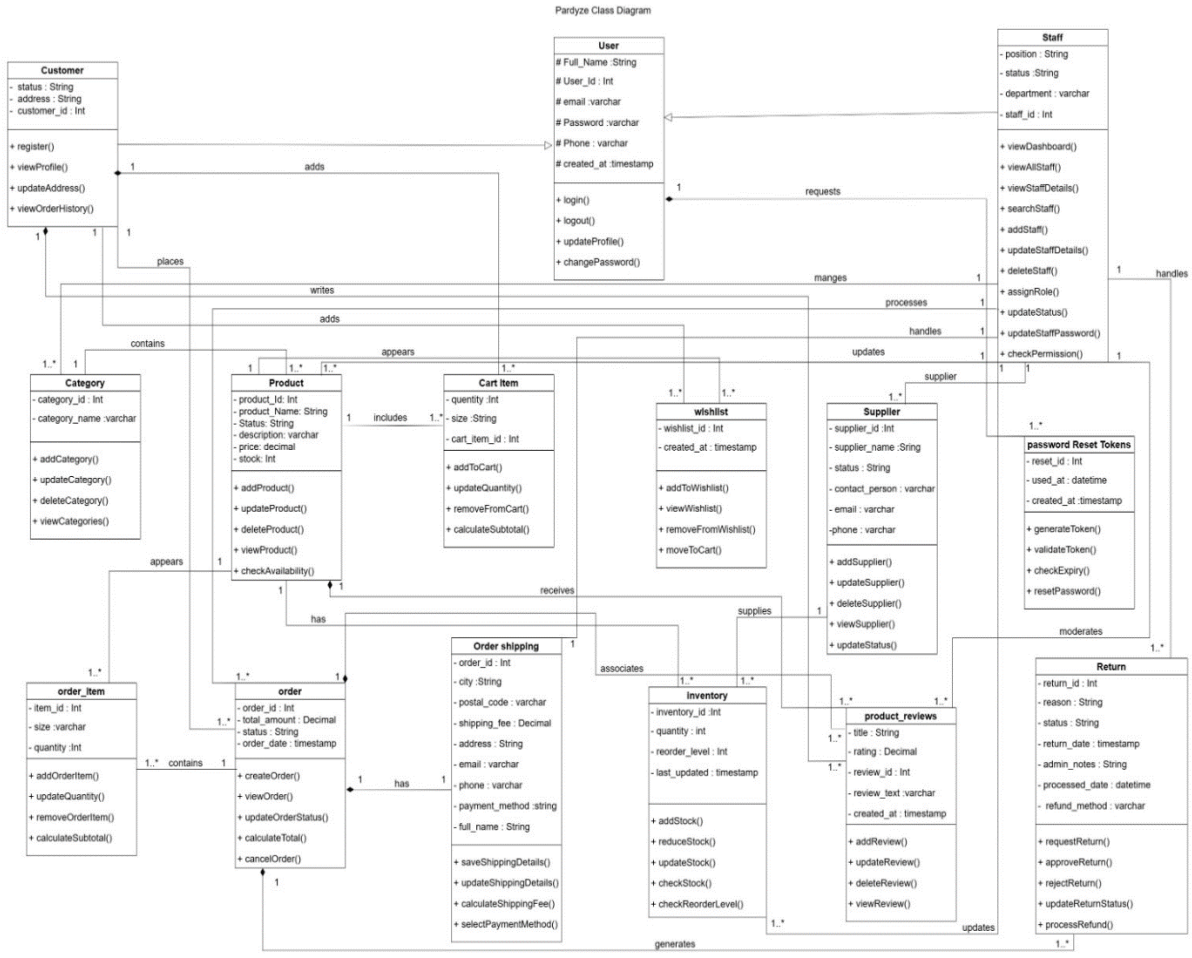


Figure 3.2

3.2.4 Sequence Diagrams of the Proposed System

Sequence diagrams represent the order in which messages are exchanged between an actor and the internal components of the system. They demonstrate how each use case is completed from the initial request until the final response.

The sequence diagrams for Paradyze generally contain the following participants:

- Actor
- User Interface
- Controller
- Service or Business Logic
- Database
- External Service, where applicable

The Actor initiates an action through an interface. The interface sends the request to a controller. The controller validates the request and communicates with the relevant service or database. The result is then returned to the interface and displayed to the actor.

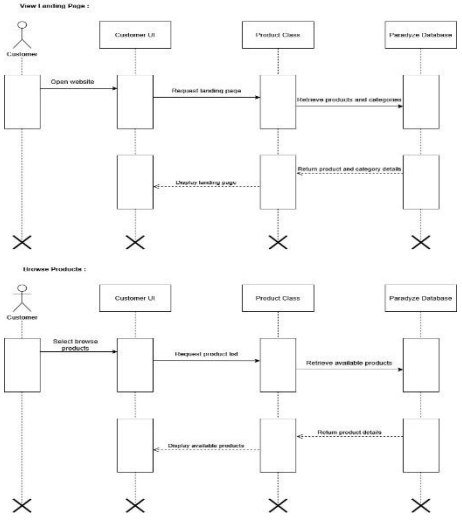


Figure 3.3

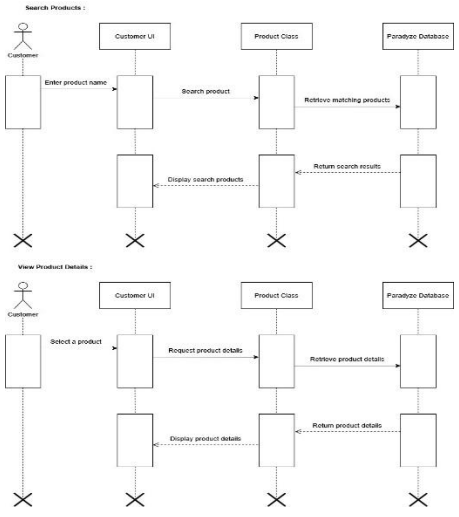


Figure 3.4

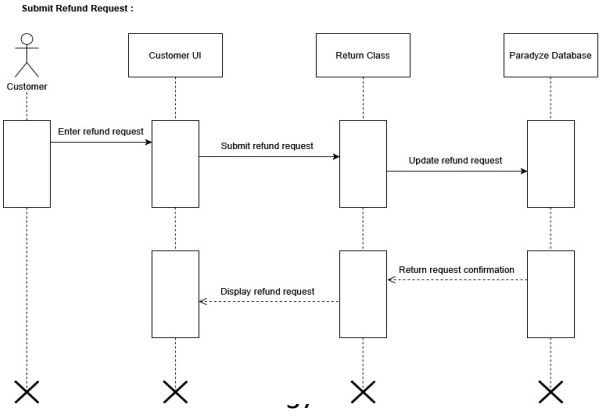


Figure 3.5

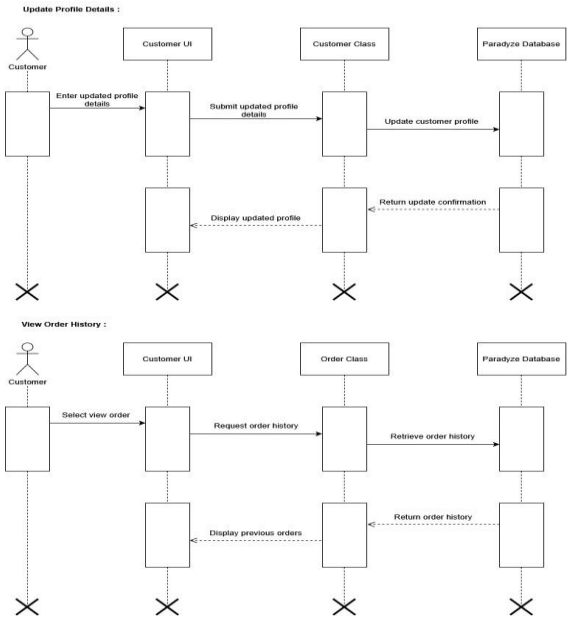


Figure 3.6

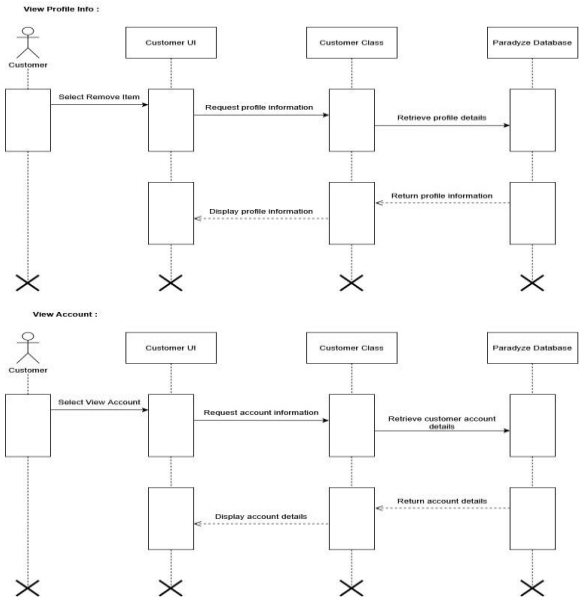


Figure 3.7

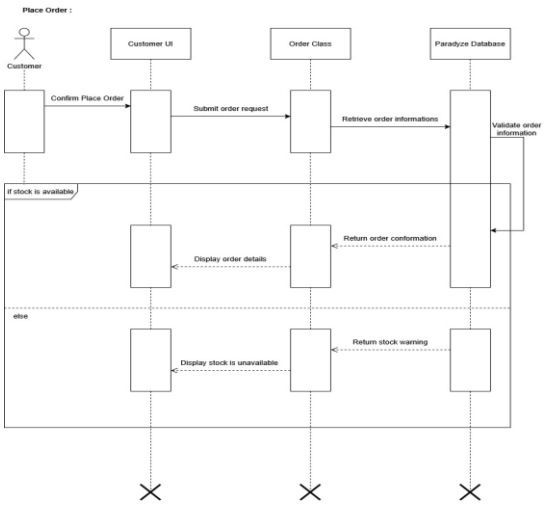


Figure 3.8

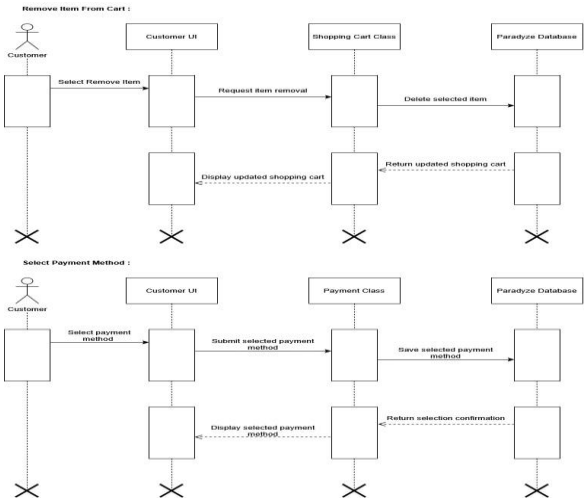


Figure 3.9

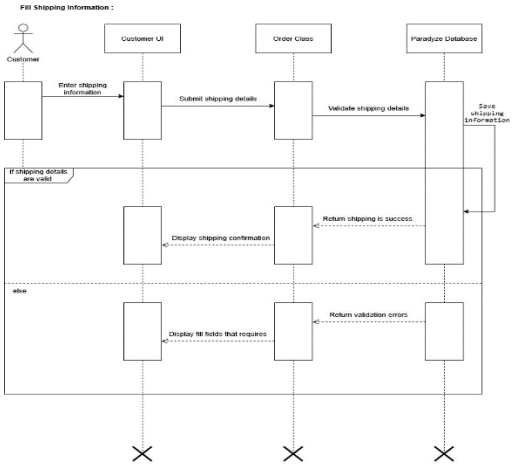


Figure 3.10

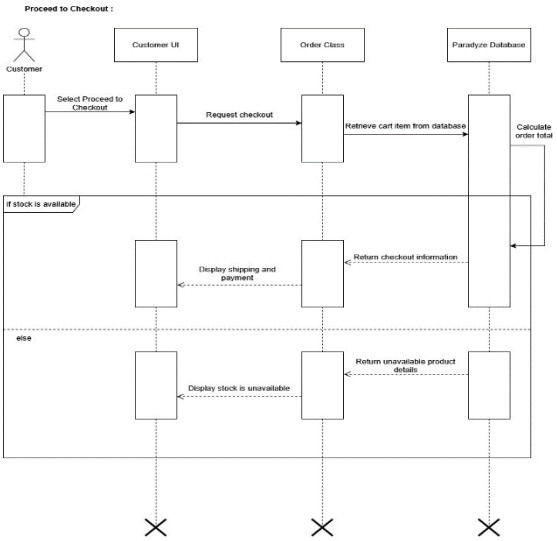


Figure 3.11

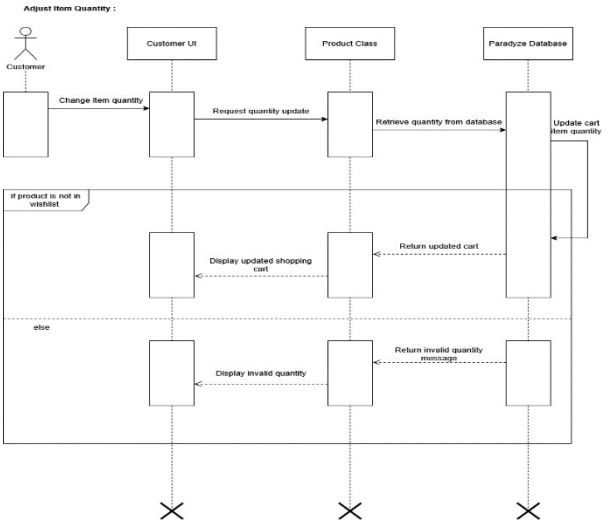


Figure 3.12

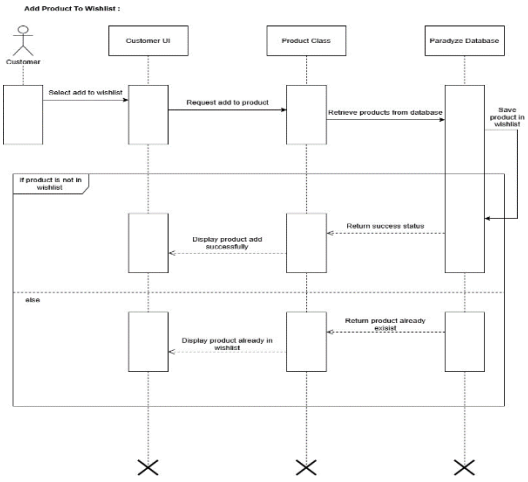


Figure 3.13

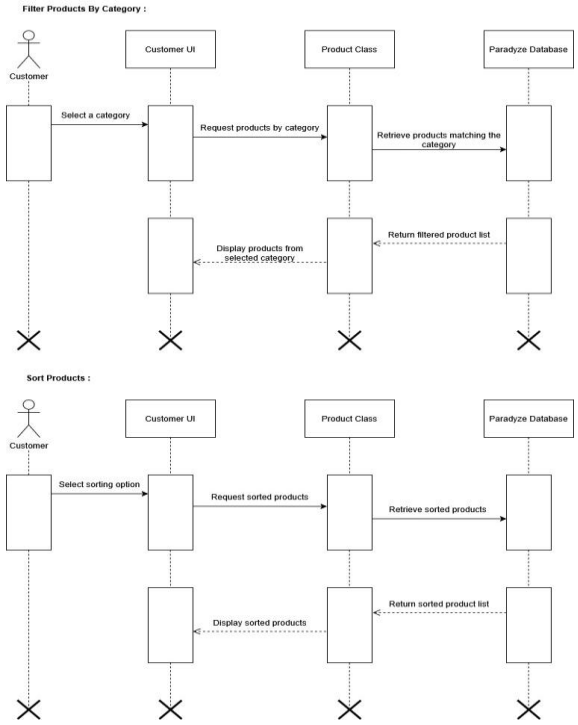


Figure 3.14

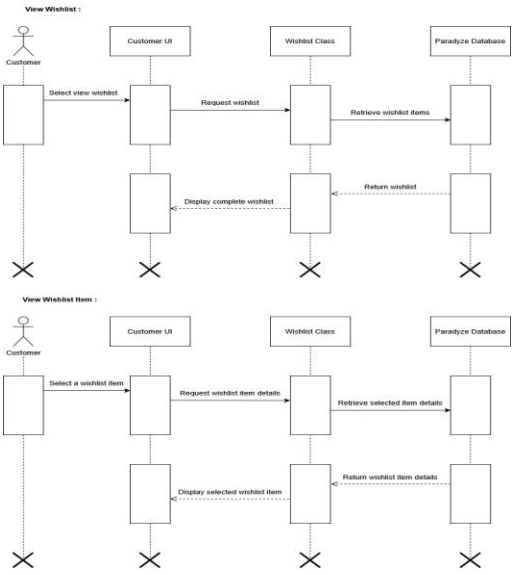


Figure 3.15

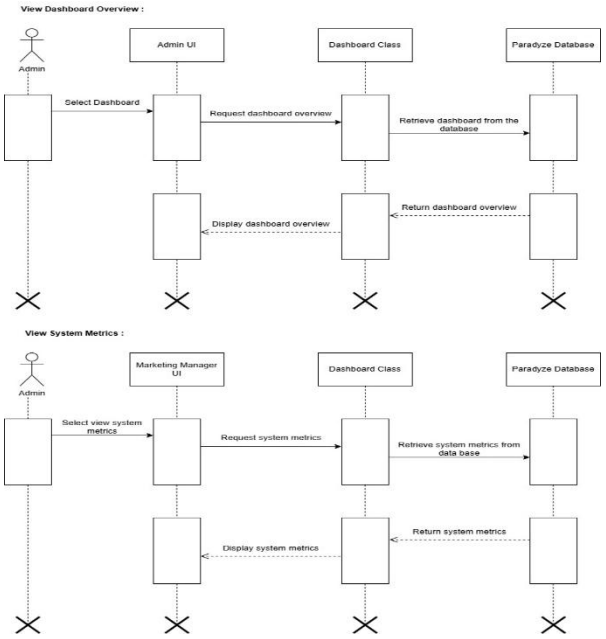


Figure 3.16

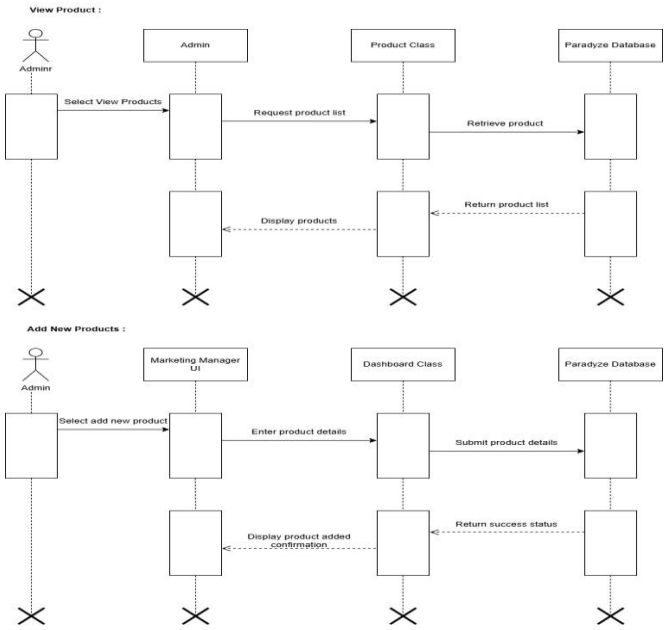


Figure 3.17

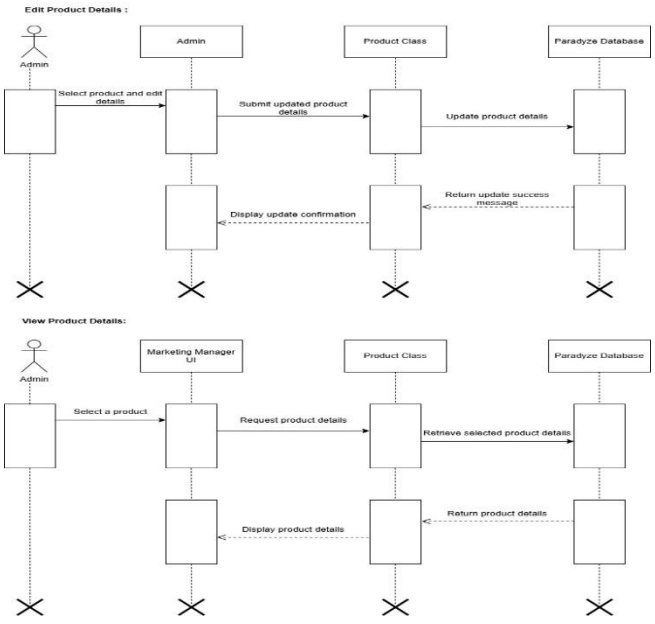


Figure 3.18

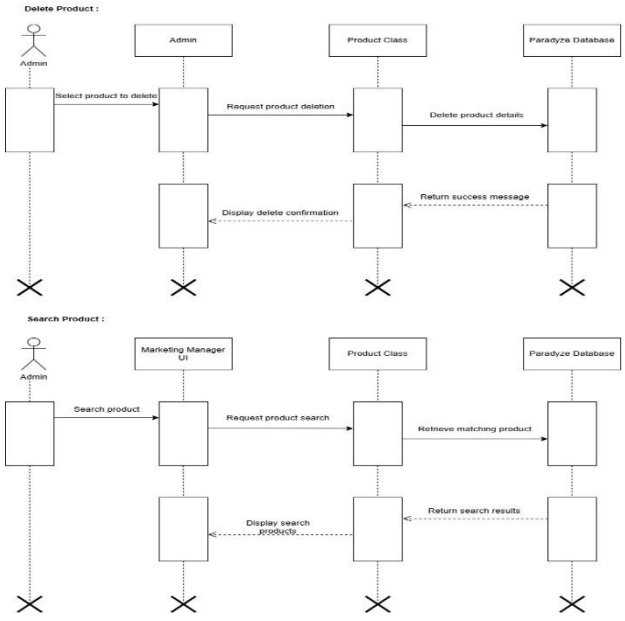


Figure 3.19

3.3 Entity Relationship Diagram of the Proposed System

The Entity Relationship Diagram represents the logical data structure of the proposed Paradyze Online Clothing Store Management System. It identifies the database entities, their main attributes, primary keys, foreign keys and relationships.

The ER diagram was developed by analyzing the information required for customer management, employee management, product management, inventory control, shopping carts, orders, payments, suppliers, shipping, returns, refunds, product reviews and reporting.

Unlike the class diagram, the ER diagram contains data attributes and database relationships. Methods and system operations are not included in the ER diagram.

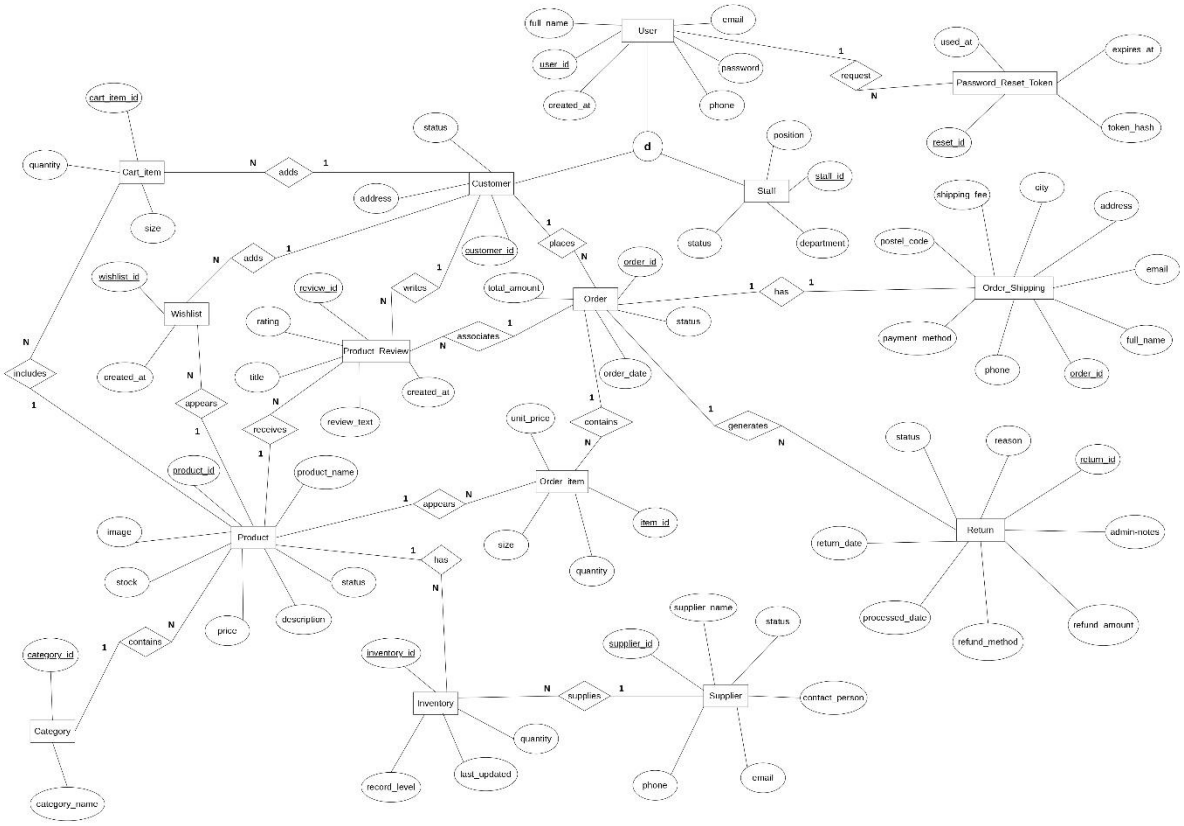


Figure 3.20

3.4 Chapter Summary

This chapter presented the analysis of the current and proposed systems of Paradyze (Private) Limited. The current system analysis showed that the organization relies heavily on Instagram, WhatsApp, spreadsheets, manual payment verification and manual delivery coordination. These practices create difficulties in maintaining accurate inventory, processing orders, storing customer details, tracking deliveries and producing reliable reports.

The proposed system uses case diagrams identified the main actors and functions of the computerized solution. Customers are provided with facilities for account management, product browsing, shopping-cart management, order placement, payment, order tracking, wishlists, reviews, returns and refunds. Staff functions are controlled through role-based access for administrators, operations managers, marketing managers, store managers and warehouse staff.

The class diagram described the static structure of the proposed solution and showed the responsibilities and relationships of classes such as User, Customer, Staff, Product, Inventory, Order, Payment, Supplier, Shipping, Return and Refund. The sequence diagrams demonstrated the interactions between actors, interfaces, controllers, services, databases and external services for the main use cases.

Finally, the Entity Relationship Diagram represented the proposed database entities, primary keys, foreign keys, relationships and cardinalities. The analysis models developed in this chapter provide the foundation for the interface, database and report designs presented in the next chapter.

Chapter 4: Solution Design

4.1 Introduction

Solution design is the process of converting the requirements identified during system analysis into a detailed plan for system implementation. It defines how users interact with the proposed solution, how information is stored and processed, and how system-generated reports are presented.

This chapter describes the design of the Paradyze Online Clothing Store Management System. The proposed system was designed as a centralized web-based platform that supports both customer shopping activities and internal business operations. Customers are provided with interfaces for account registration, product browsing, cart management, checkout, order tracking, wishlist management and product reviews. Staff members are provided with role-based interfaces for managing products, customers, inventory, orders, suppliers, shipping, returns, refunds and reports.

The design was prepared by considering the requirements identified through interviews, observations, questionnaires and document reviews. The existing Paradyze operation depends mainly on Instagram, WhatsApp and manually maintained spreadsheets. Therefore, the main purpose of the proposed design is to reduce manual work and provide an integrated system for managing the organization's online clothing business.

This chapter contains the following main design areas:

- Interface design
- Database design

4.2 Interface Design

Interface design determines how users communicate with the proposed system. A successful interface should be easy to understand, visually attractive, responsive, consistent and suitable for the responsibilities of each user.

The interfaces of the Paradyze system were designed using the organization's branding style. A dark green color theme, simple navigation, product images and consistent buttons were used throughout the system. Customer interfaces were designed to provide a smooth shopping experience, while administrative interfaces were designed to support efficient business management.

The following design principles were considered:

- Clear and consistent navigation
- Simple page layouts
- Responsive design for desktops, tablets and mobile devices
- Readable fonts and suitable text sizes
- Consistent buttons, input fields and icons
- Appropriate validation and error messages
- Easy access to frequently used functions
- Role-based access to staff interfaces
- Confirmation messages for important actions
- Protection against unauthorized access

Each interface is presented with an interface number, interface name and description.

4.2.1 Customer Interface Designs

Interface UI-01: Customer Home Page

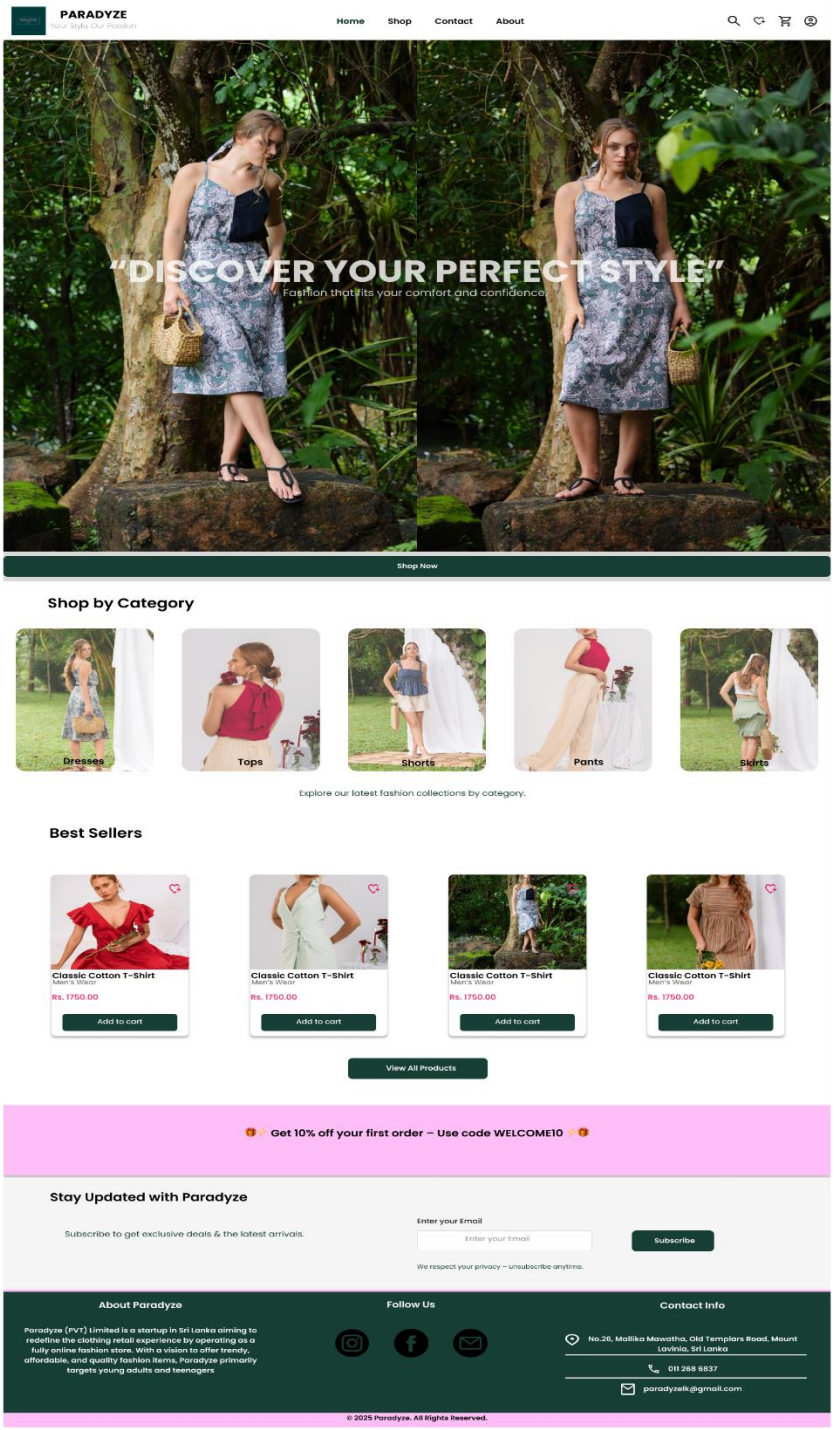


Figure 4.1

Interface UI-02: Customer Shop Page

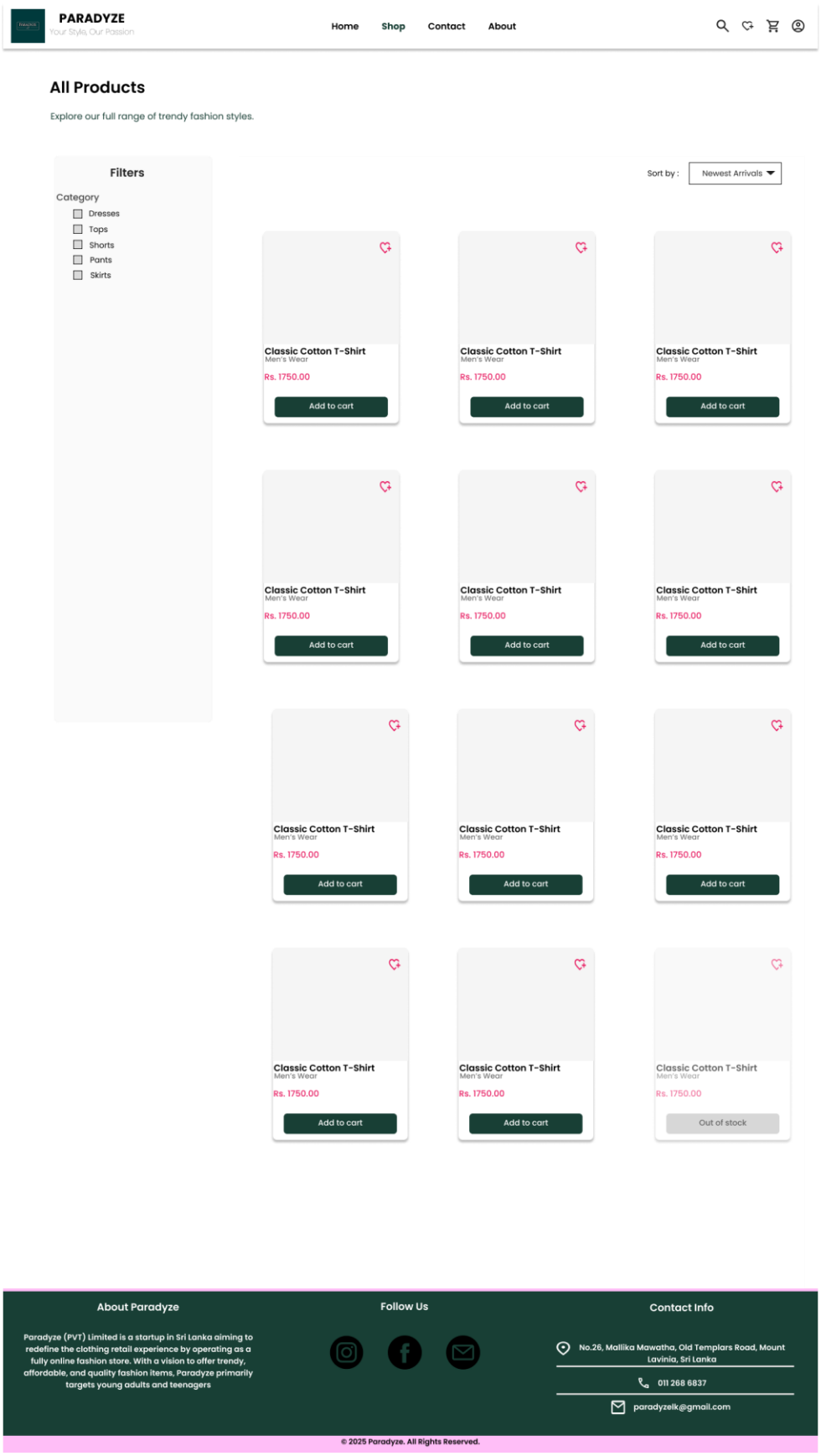


Figure 4.2

Interface UI-03: Customer Product Page

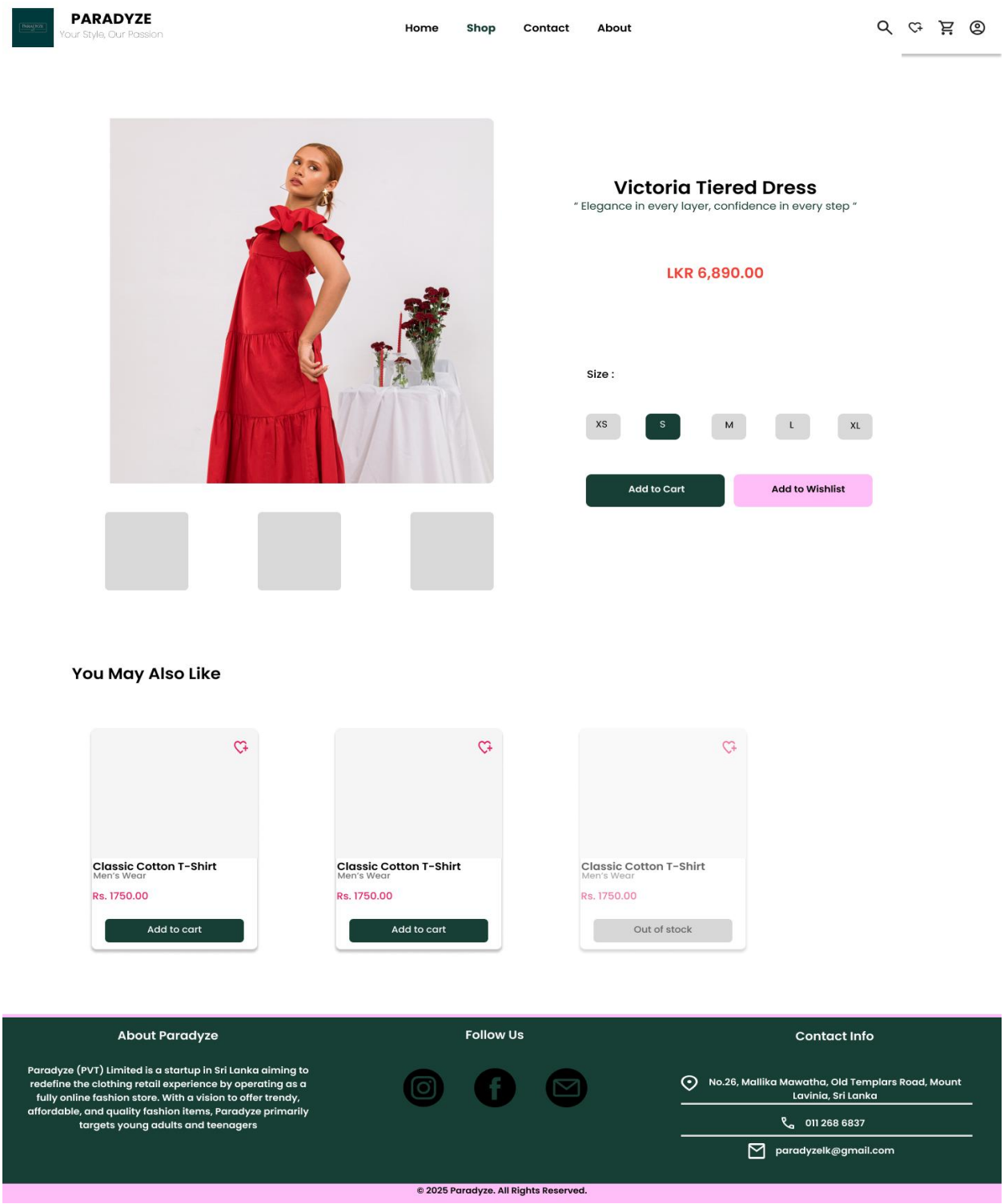


Figure 4.3

Interface UI-04: Customer Cart Page

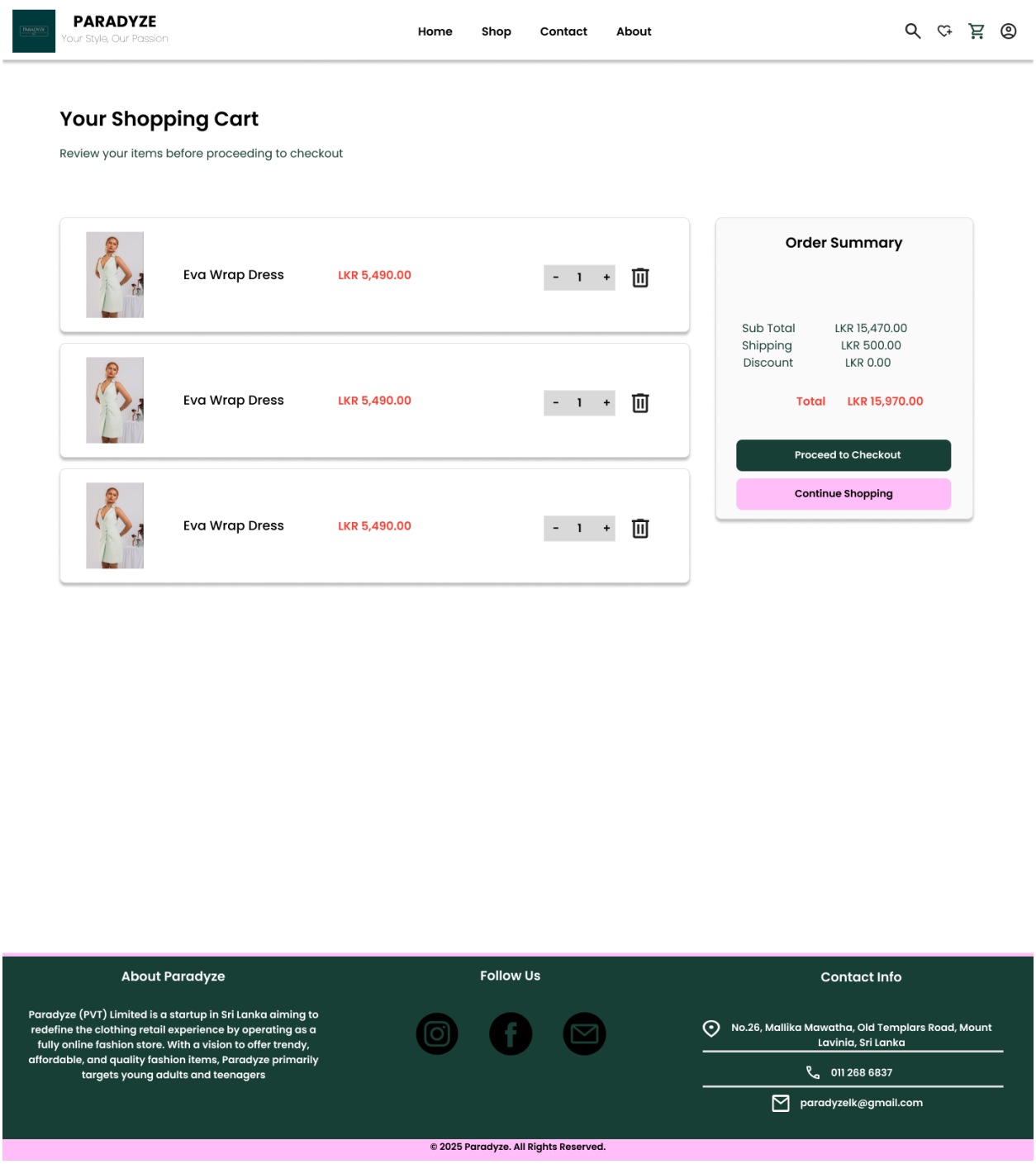


Figure 4.4

Interface UI-05: Customer Checkout Page

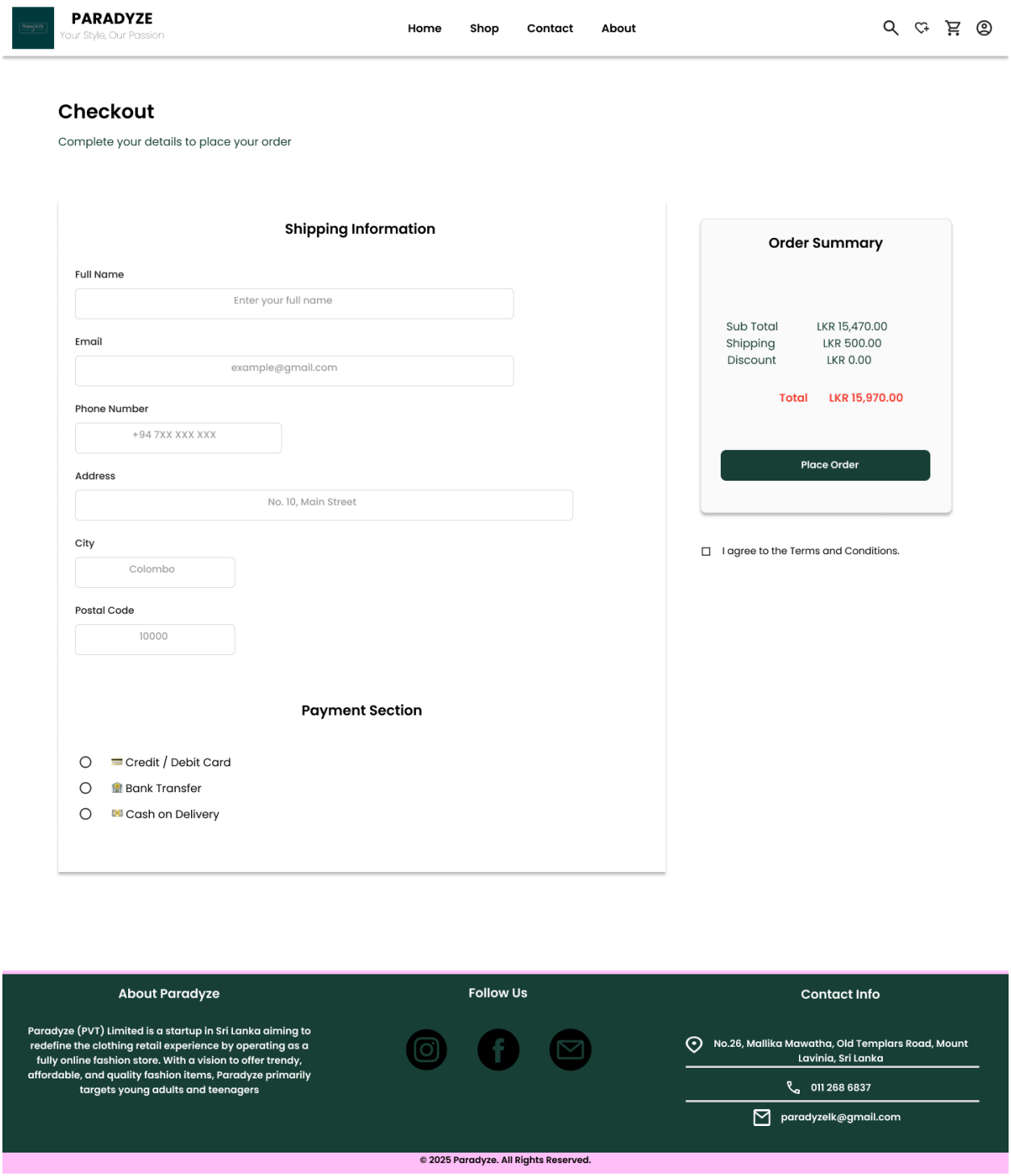


Figure 4.5

Interface UI-06: Customer Order Confirmation Page

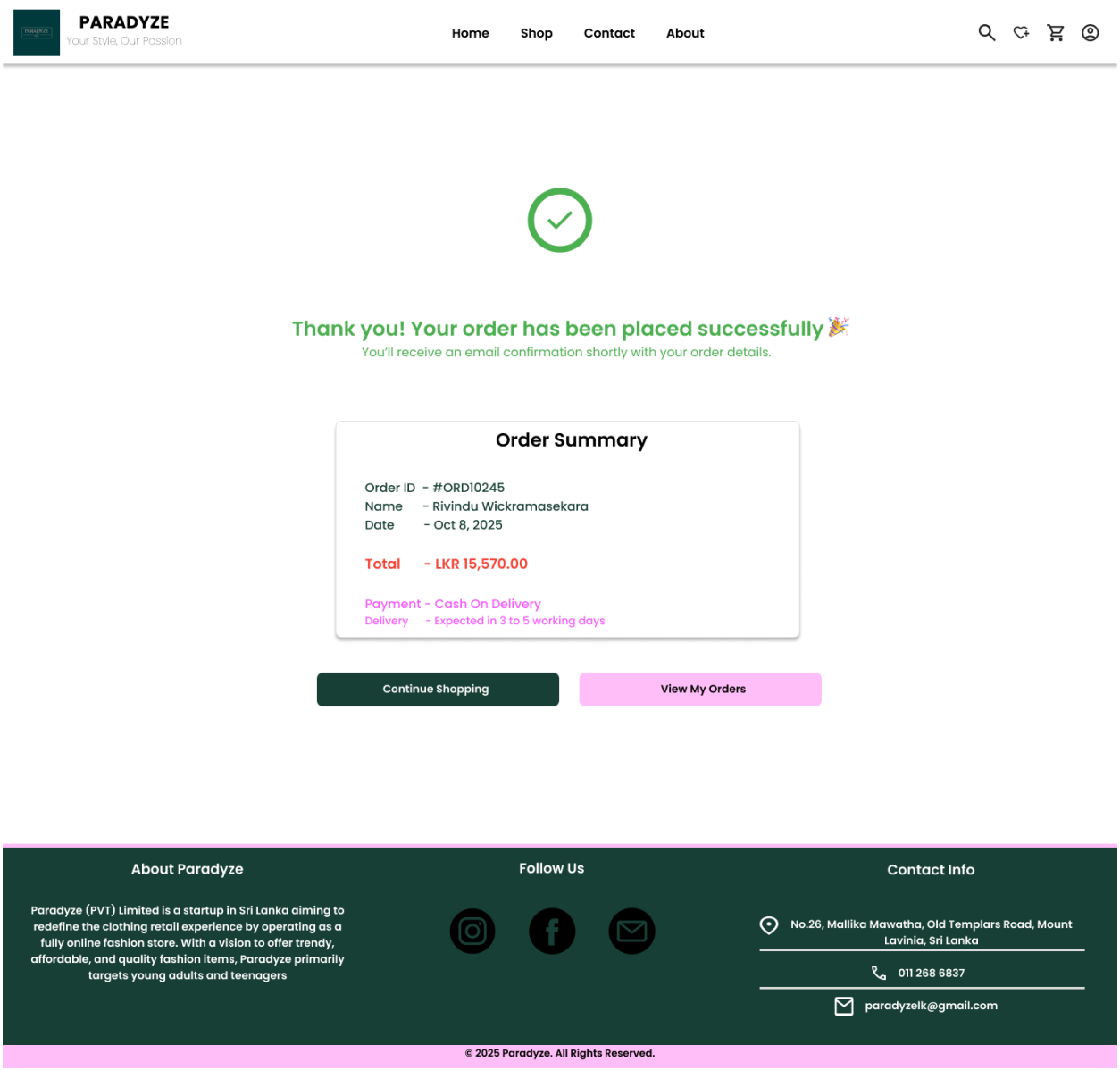


Figure 4.6

Interface UI-07: Customer Wishlist Page

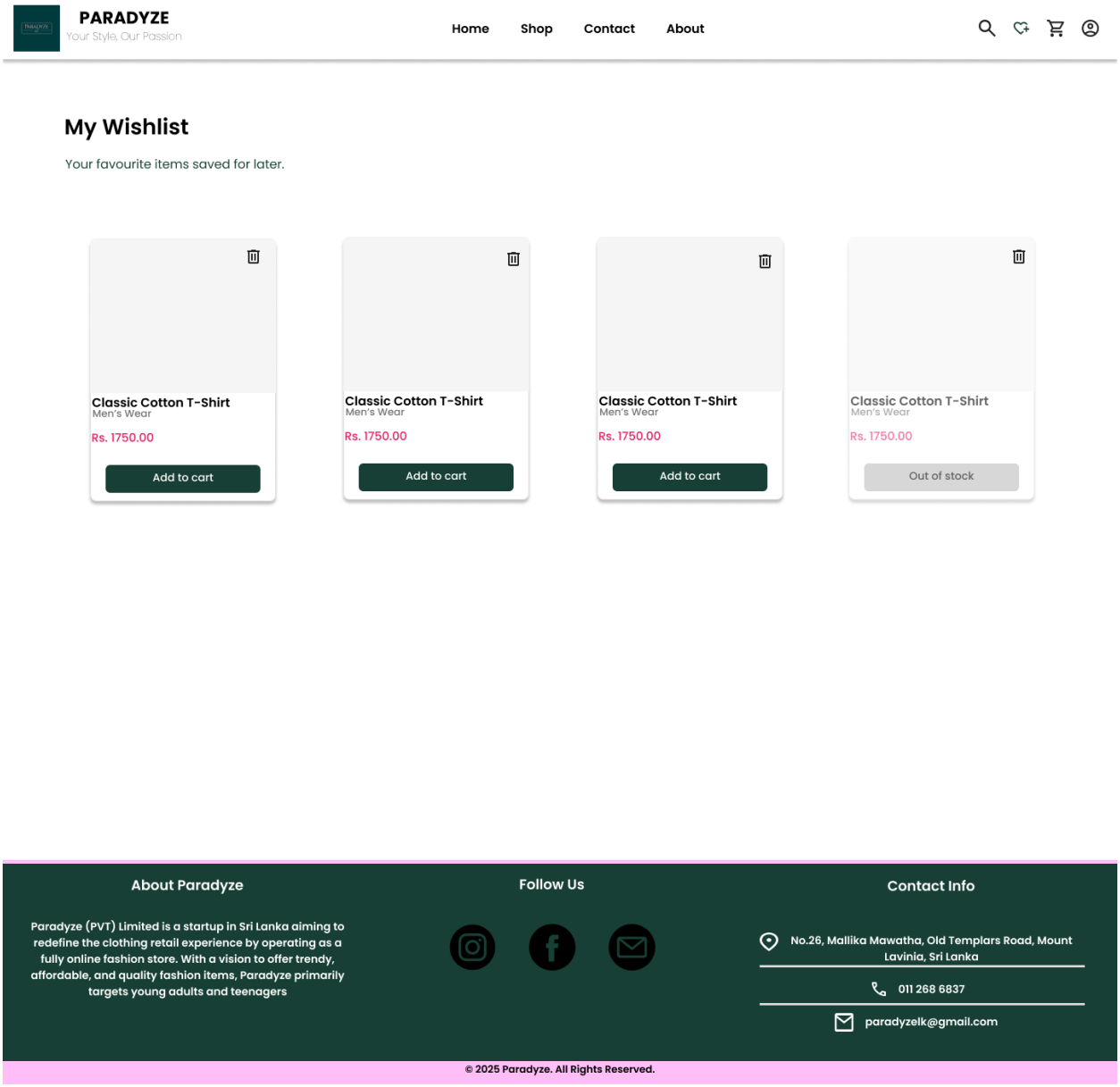


Figure 4.7

Interface UI-08: Customer Wishlist – Empty State Page

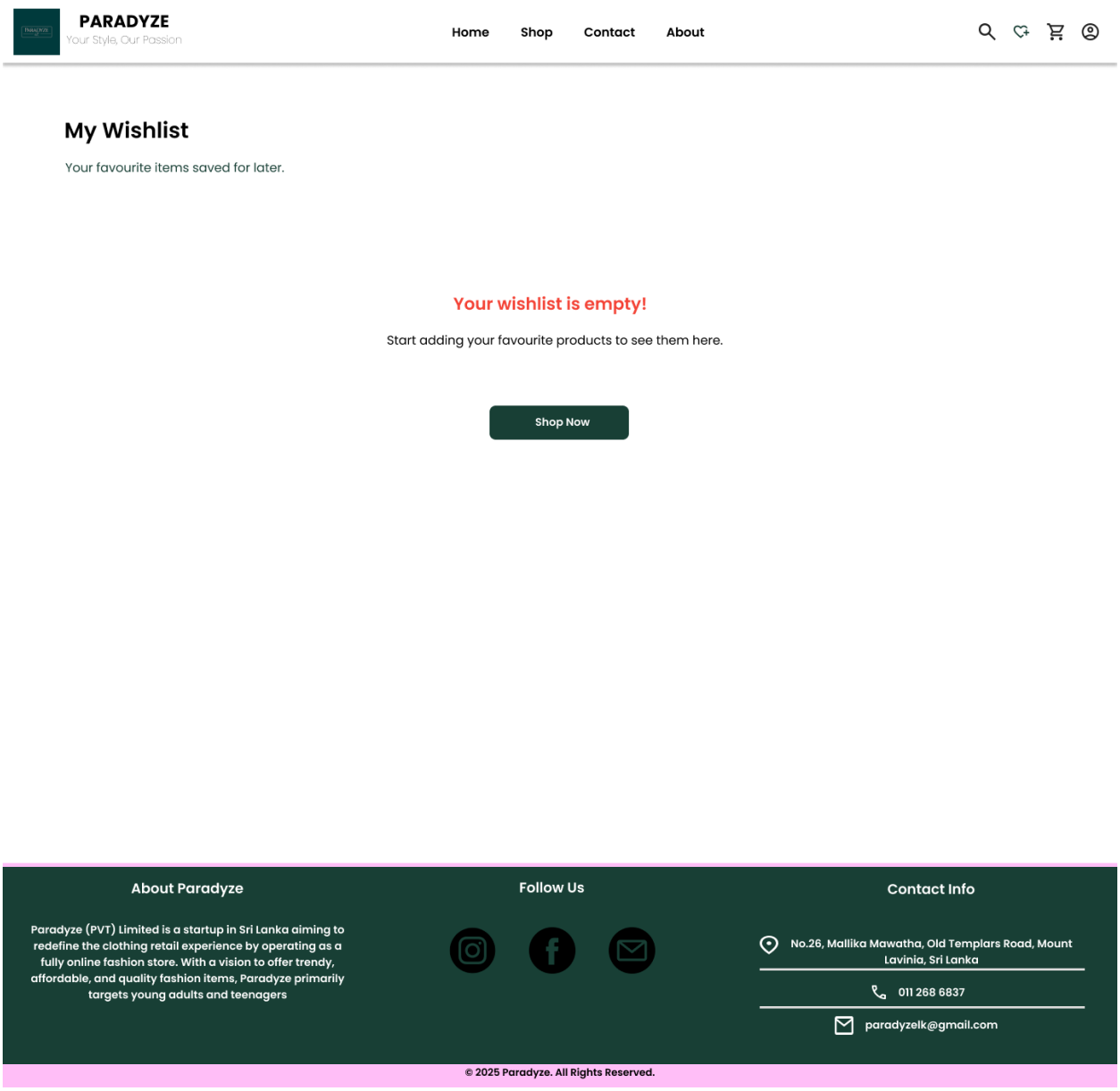


Figure 4.8

Interface UI-09: User Authentication Page

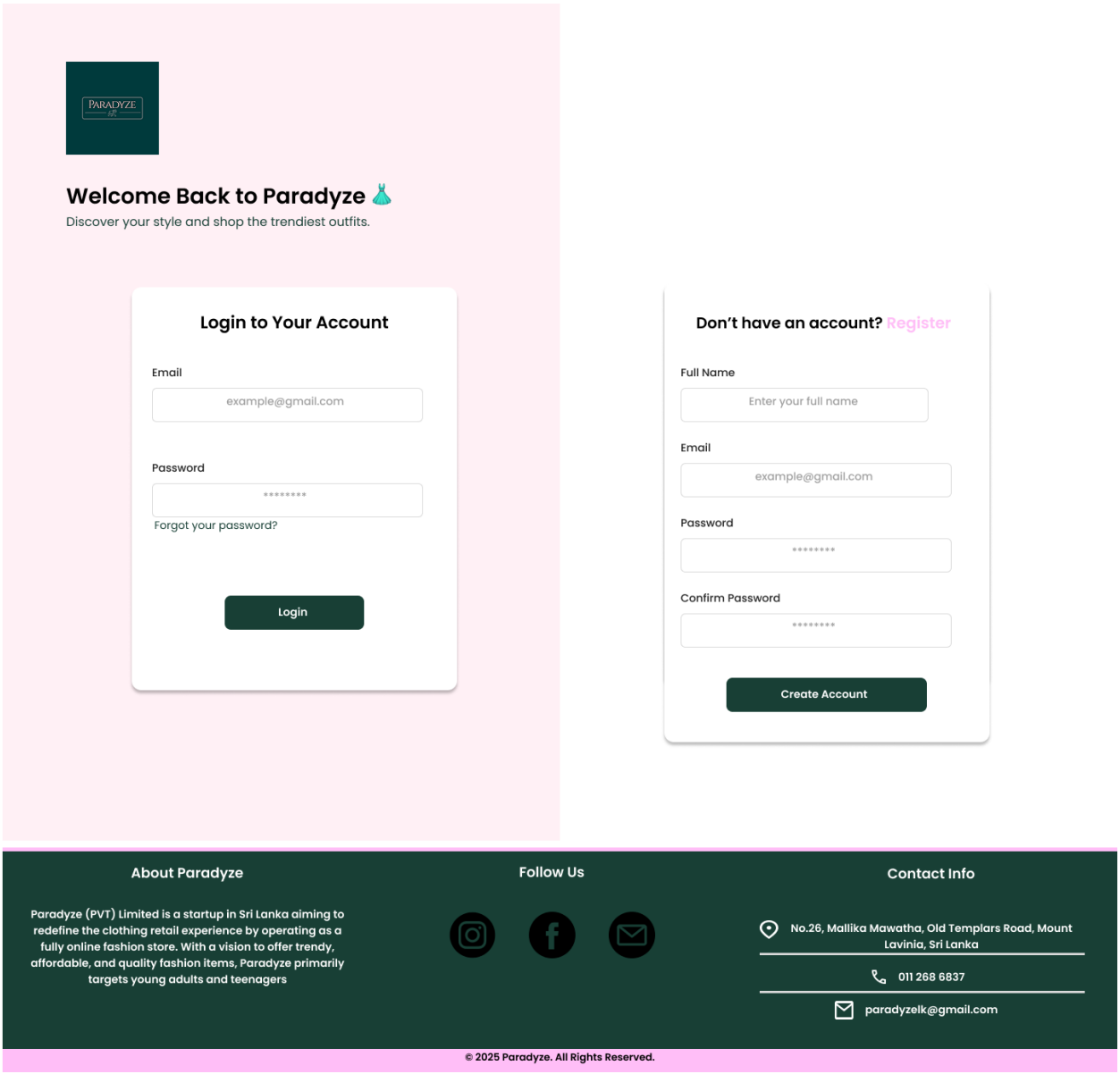


Figure 4.9

Interface UI-10: My Account – Profile Info Page

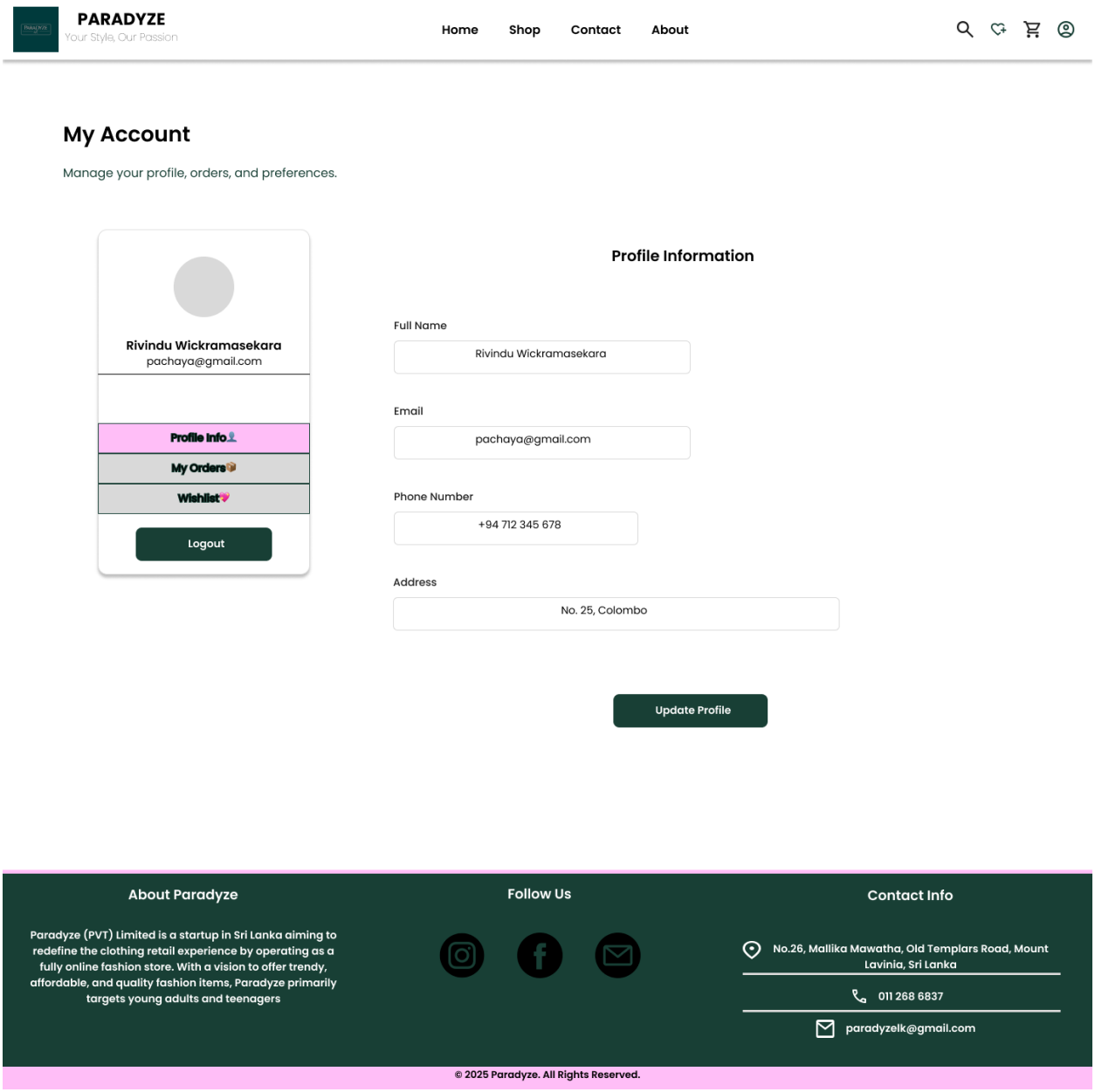


Figure 4.10

Interface UI-11: My Account – Orders Page

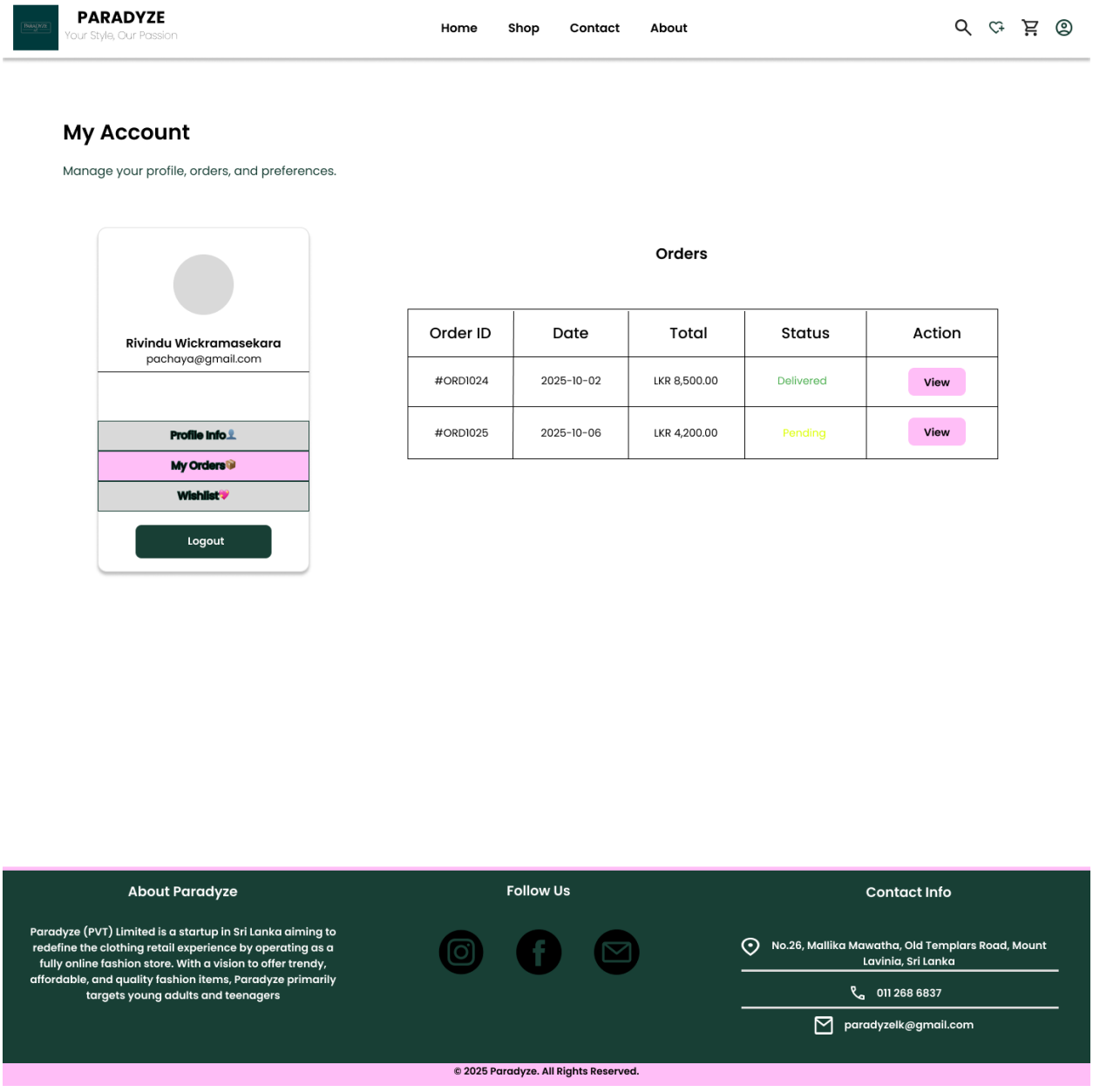


Figure 4.11

Interface UI-12: My Account – Wishlist Page

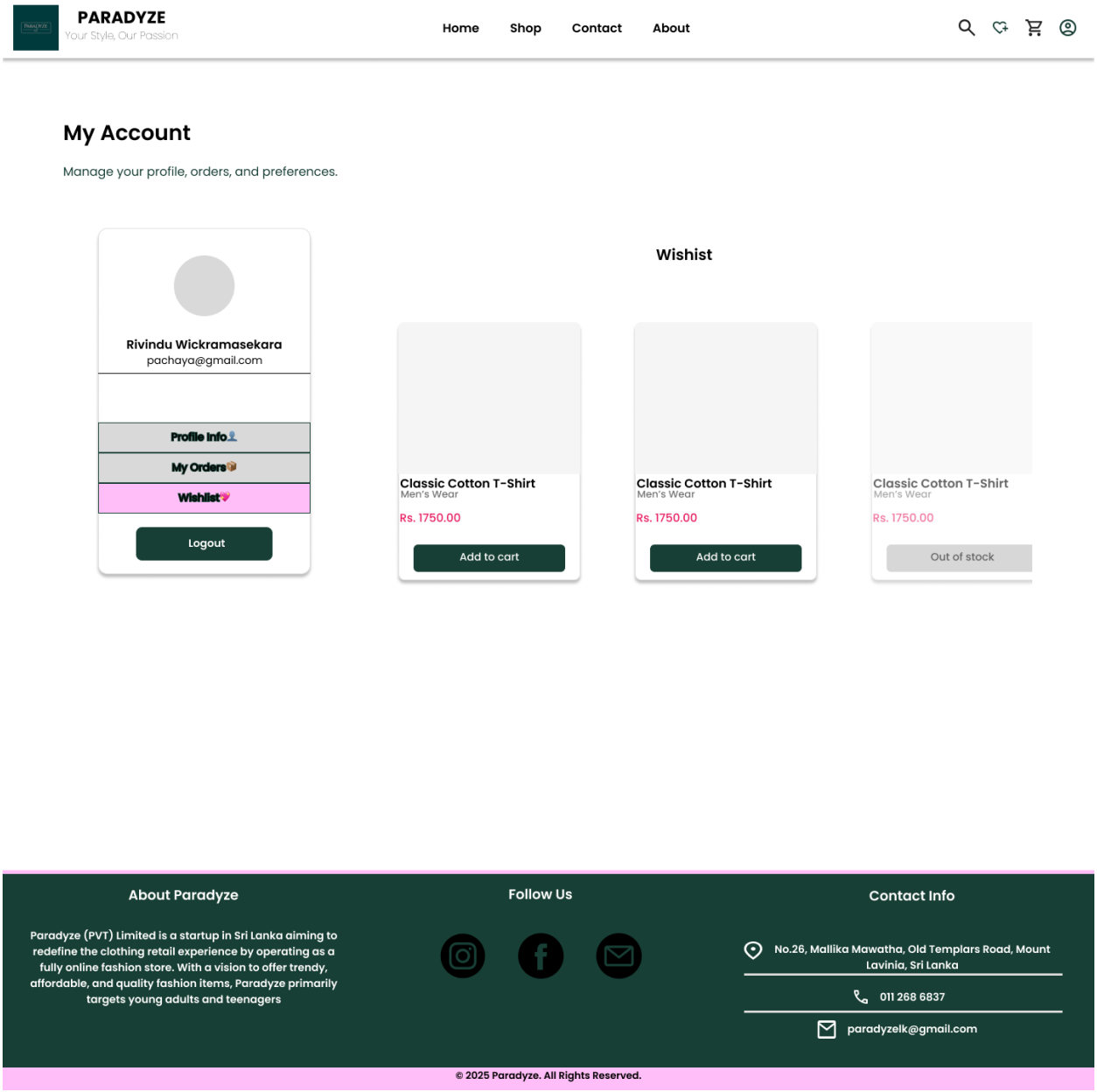


Figure 4.12

4.2.2 Staff and Administrative Interface Designs

Interface UI-13: Admin Dashboard Page

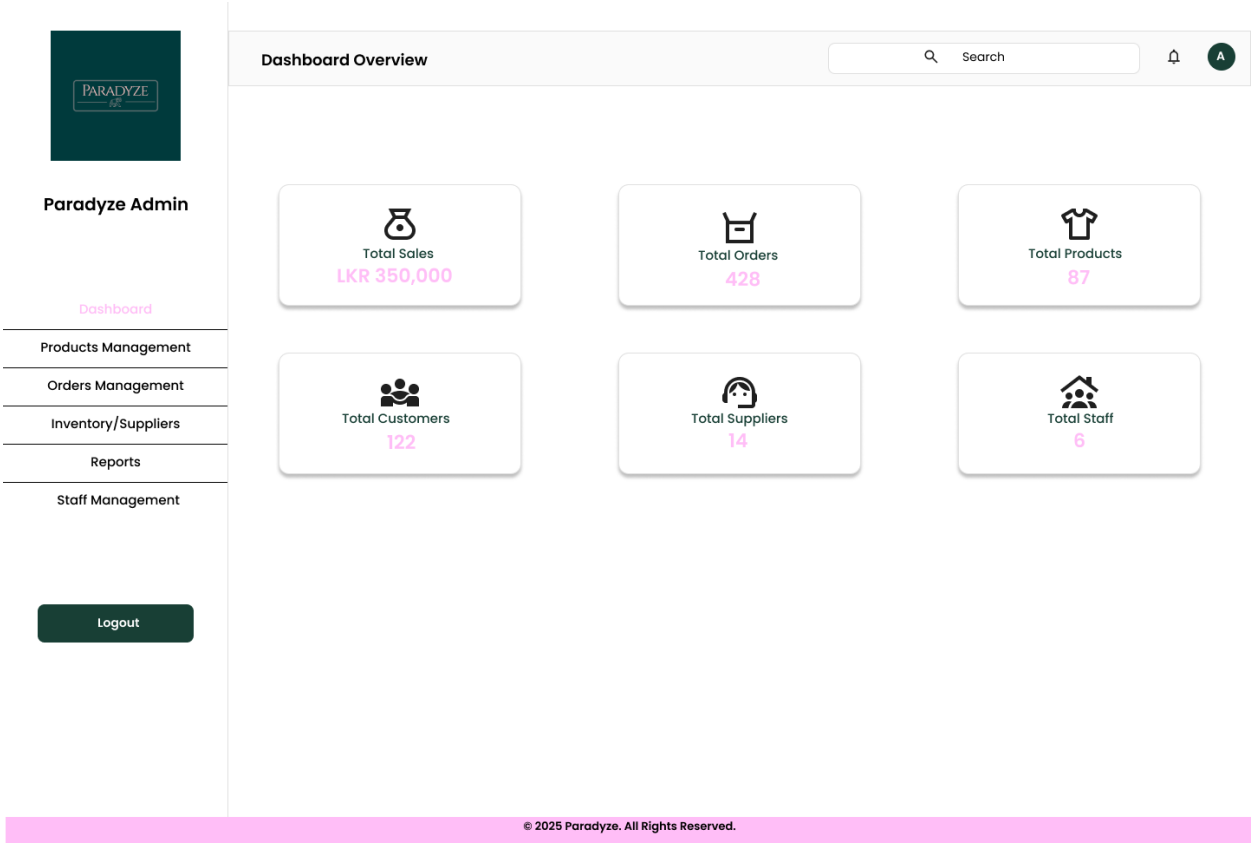


Figure 4.13

Interface UI-14: Product Management Page

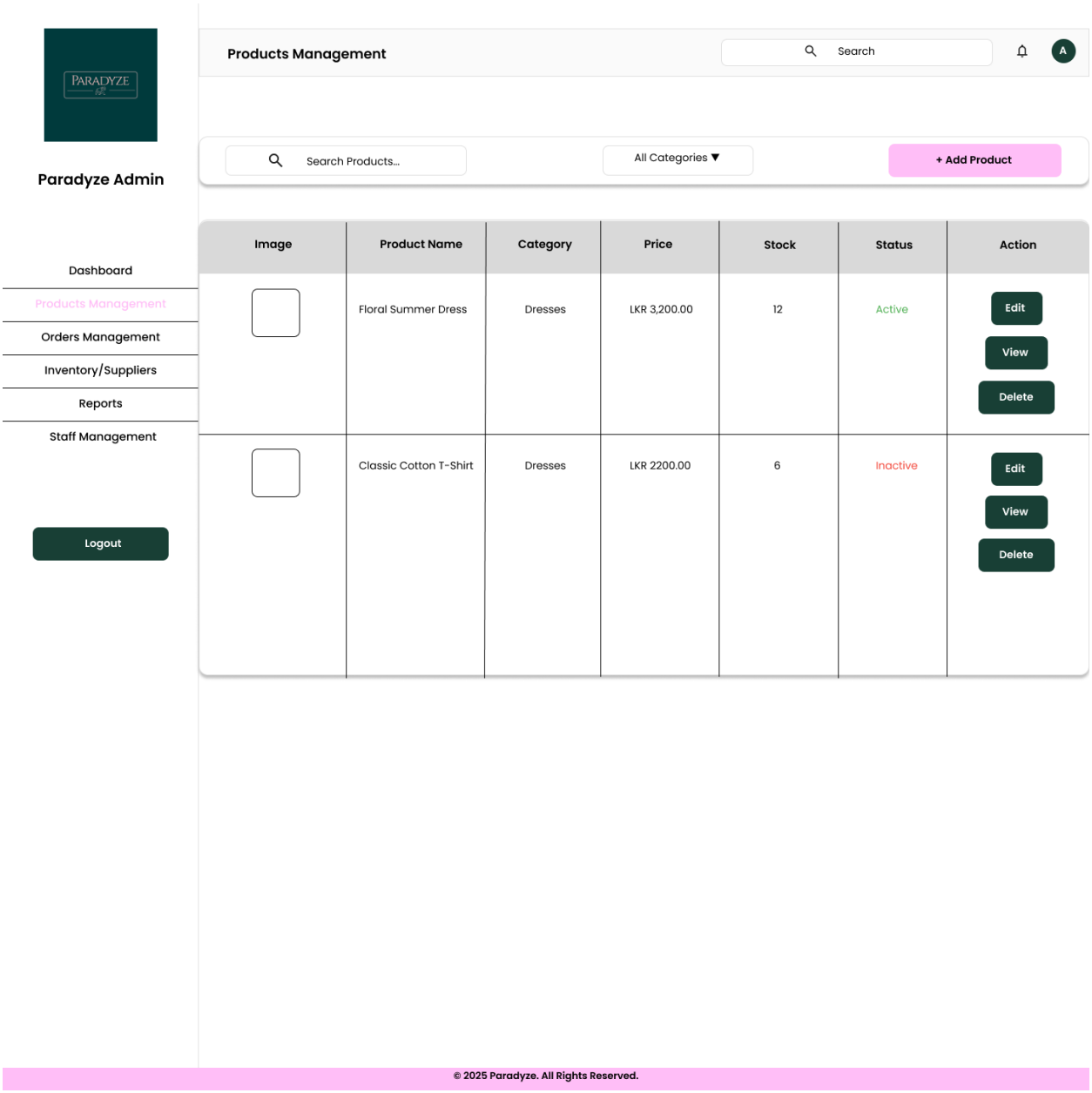


Figure 4.14

Interface UI-15: Product Management Page

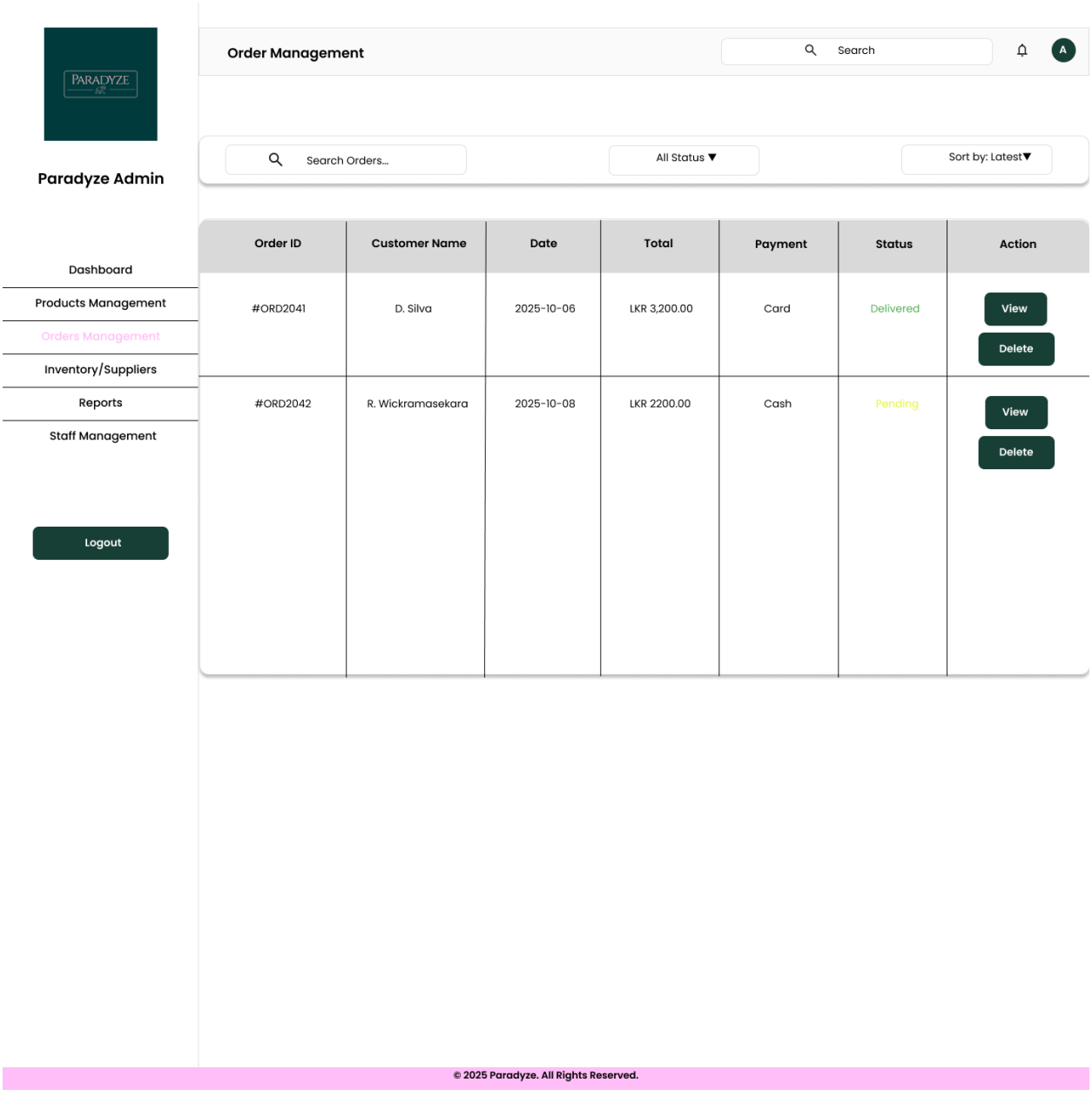


Figure 4.15

Interface UI-16: Reports Page

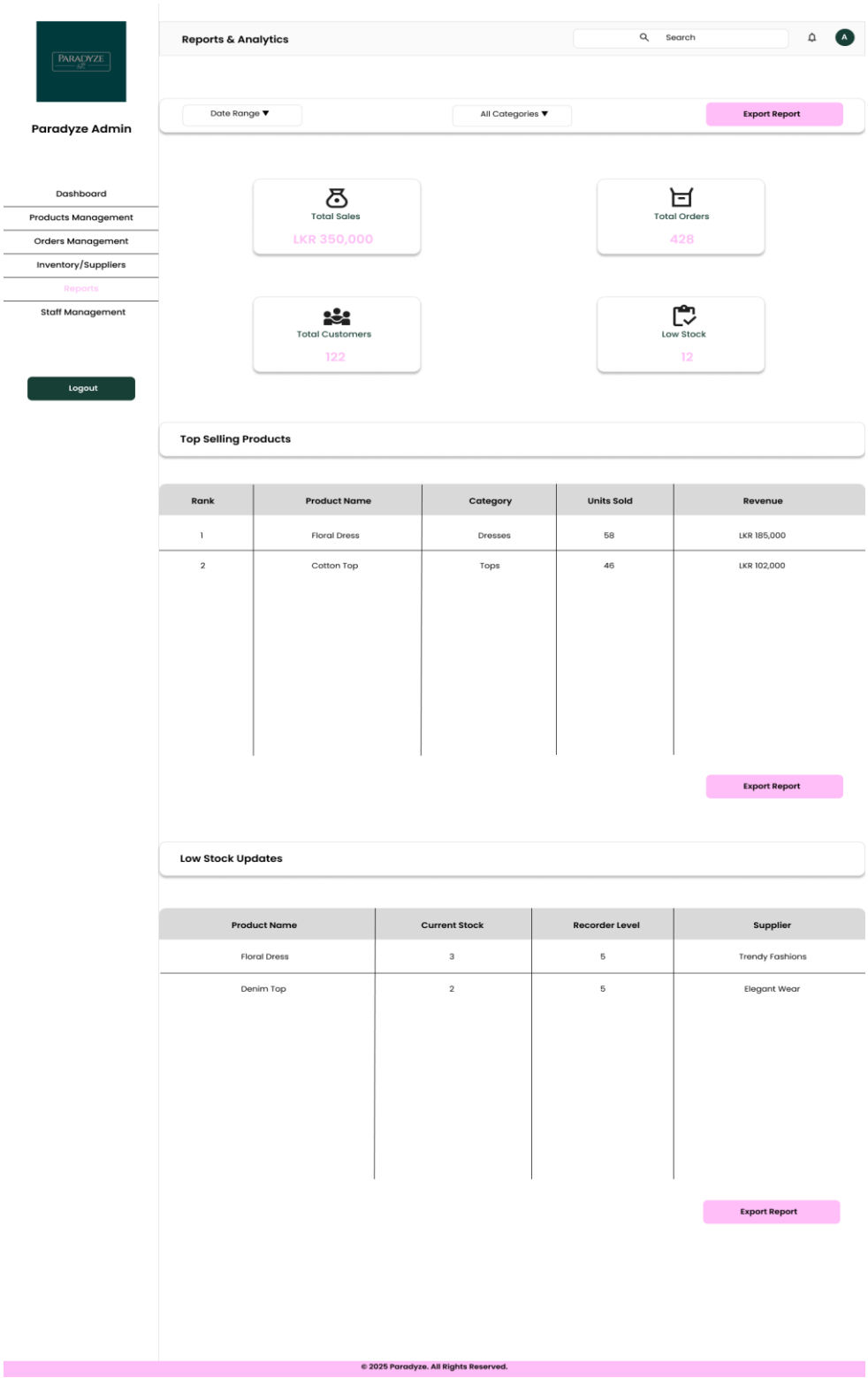


Figure 4.16

Interface UI-17: Staff Management Page

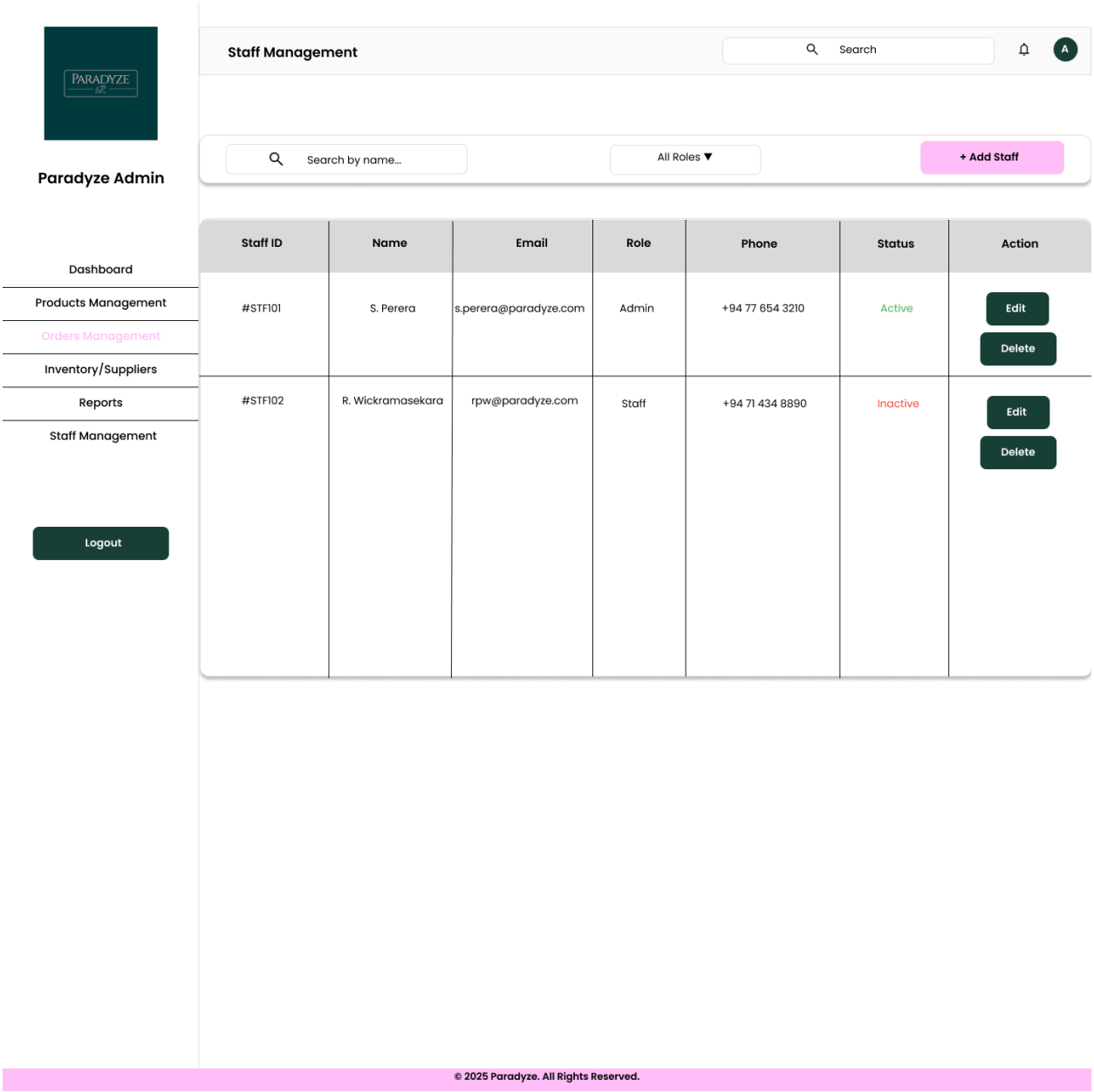


Figure 4.17

4.3 Database Design

Database design defines how the system's data is organized, stored and related. The proposed database was designed to support customer accounts, staff accounts, products, inventory, carts, wishlists, orders, payments, suppliers, shipping, reviews, returns, refunds and reporting activities.

A relational database model was selected because it supports structured data, relationships, integrity constraints and efficient data retrieval. MySQL was selected as the database management system.

The database design follows the following principles:

- Each table has a primary key.
- Foreign keys connect related tables.
- Repeated data is minimized.
- User authentication data is stored centrally.
- Customer and staff details are separated from common user details.
- Historical order information is preserved.
- Products are not permanently deleted where historical records exist.
- Referential integrity is maintained.
- Sensitive information, such as passwords, is stored securely.
- Data types and constraints are used to prevent invalid records.

4.3.1 Database Table Summary

Table No.	Table Name	Primary Key	Main Foreign Keys	Description
DB-01	User	user_id	None	Stores common authentication and personal details for customers and staff.
DB-02	Customer	customer_id	user_id	Stores customer-specific details.
DB-03	Staff	staff_id	user_id	Stores staff-specific and employment details.
DB-04	Category	category_id	None	Stores product categories.
DB-05	Product	product_id	category_id	Stores product information.
DB-06	Supplier	supplier_id	None	Stores supplier details.
DB-07	Inventory	inventory_id	product_id, supplier_id	Stores stock quantities and reorder information.
DB-08	Order	order_id	customer_id	Stores order-header information.
DB-09	Order_Items	item_id	order_id, product_id	Stores products included in an order.
DB-10	Order_Shipping	order_id	order_id	Stores delivery and shipping details.
DB-11	Return	return_id	order_id	Stores product-return requests.
DB-12	Cart_Item	cart_item_id	customer_id, product_id	Stores products saved in cart.
DB-13	Wishlist	wishlist_id	customer_id, product_id	Stores wishlist items.
DB-14	Product_Reviews	review_id	customer_id, product_id, order_id	Stores product ratings and reviews.
DB-15	Password_Reset_ Token	reset_id	user_id	Stores password reset details.

Table 2

4.4 Report Layout Design

Report-layout design determines how business information is presented to users. Reports assist managers in monitoring sales, inventory, orders, customers, suppliers, returns, refunds and staff activities.

The system allows authorized users to generate reports using selected filters such as date range, product, category, customer, supplier and status. Reports may be displayed on screen and exported to PDF where implemented.



Paradyze Reports & Analytics

Total Sales: LKR 0.00

Total Orders: 0

Total Customers: 7

Rank	Product	Category	Units	Revenue
1	Summer Vacay Dress	Dresses	0	LKR 0.00
2	Emerald Chic Short	Shorts	0	LKR 0.00
3	Paisley Skirt	Skirts	0	LKR 0.00
4	Luxe Flow Wide-Leg Pants	Pants	0	LKR 0.00
5	Summer Vacay Dress	Dresses	0	LKR 0.00
6	Athena Strap Top	Tops	0	LKR 0.00
7	Full Strap Dress	Dresses	0	LKR 0.00
8	Chic Ruffle Skirt	Skirts	0	LKR 0.00
9	Summer Vacay Dress	Skirts	0	LKR 0.00
10	Victoria Tiered Dress	Dresses	0	LKR 0.00

Figure 4.18

Chapter 5: Conclusion

5.1 Introduction

This chapter summarizes the achievements, limitations and possible future improvements of the Paradyze Online Clothing Store Management System**.**

The project was developed to reduce the difficulties caused by Paradyze's manual business processes. Previously, products were promoted through Instagram and WhatsApp, while orders, customer information and inventory were mainly managed using messages and spreadsheets.

5.2 Project Achievements

The project introduced a centralized system for managing the main activities of Paradyze.

The developed system allows customers to:

- Register and log in
- Browse and search products
- Manage a shopping cart and wishlist
- Place orders
- View order history
- Track order status
- Submit reviews
- Request returns and refunds

The staff section allows authorized employees to:

- Manage products and categories
- Update inventory
- Process customer orders
- Manage customers and staff
- Handle shipping, returns and refunds
- Manage suppliers
- Generate business reports

The system also uses role-based access control so that staff members can access only the functions related to their responsibilities.

Overall, the system improves data accuracy, reduces manual work and provides customers with a more convenient shopping experience.

5.3 Project Limitations

The current system has several limitations:

- It may operate only in a local development environment.
- A real payment gateway is not fully integrated.
- Delivery tracking may require manual updates.
- Email and SMS notifications may not be fully implemented.
- The system has not been tested with a very large number of users.
- A separate mobile application is not available.
- Automated cloud backup is not implemented.

These limitations do not affect the main purpose of the prototype but should be considered before real-world deployment.

5.4 Future Enhancements

The following features can be introduced in future versions:

- Real online payment integration
- Courier and live delivery tracking
- Mobile application
- Email and SMS notifications
- Cloud deployment and automated backups
- Sinhala and Tamil language support
- Discount and loyalty programs
- AI-based product recommendations
- Barcode-based inventory management
- Advanced sales and customer analytics

5.5 Final Conclusion

The Paradyze Online Clothing Store Management System successfully provides a centralized solution for managing customers, products, inventory, orders, suppliers, shipping, returns and reports.

The system reduces the organization's dependency on social-media messages and manually maintained spreadsheets. It also improves inventory accuracy, order processing and access to business information.

Although further improvements are required before commercial deployment, the project has achieved its main objectives and provides a strong foundation for the future growth of Paradyze's online clothing business.

Appendices

- **Letter Issued by the organization:** (Placeholder for the official letter from Paradyze).
- **Meeting Minutes:** (Placeholder for records of team and stakeholder meetings).
- **Log Sheet:** (Placeholder for development hours/task tracking)
- Use Case Diagram
- Class Diagram
- ER Diagram
- Sequence Diagrams for each use-case